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This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	12 Apr 2016	First Release	MUHAMMED SHAMEEM
01	2 Oct 2016	Template update & Content update	ONG KEAN SHENG
02	19 July 2017	Content Update	MUHAMMED SHAMEEM

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Pre-Requisites

1. V810 Software requirement. Refer to Table 1 for the software requirement for each system type.

System Type	Software requirement
Standard V810	5.8 & above
XXL V810	5.8 & above
V810 S2EX	5.8 & above

Table 1 Software requirement

2. May refer to Table 2 for adjustment method.

System Type	Machine Serial Number	Adjustment method	Tuning Procedure
Standard V810	MY10133110 & below	Manual Load Jig	Automated PSP 2 Adjustment Procedure (Manual Load Jig)
	MY10133111 & above	On Rail Jig	Automated PSP 2 Adjustment Procedure (On Rail Jig)
XXL V810	All	On Rail Jig	Automated XXL PSP 2 Adjustment Procedure (On Rail Jig)
V810 S2EX	All	On Rail Jig	Automated S2EX PSP 2 Adjustment Procedure (On Rail Jig)

Table 2 Adjustment Method

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3. Prepare the On-Rail Jig as shown below in Figure 1 and Figure 2.



Figure 1 Front View



Figure 2 Rear View

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Remarks: This procedure is only applicable for the machine MY10133110 and below.

1. Run 1 loop of full Confirmation, Diagnostics and Adjustment (CD&A) test and close the V810 software.
2. Go to the following directory **C:\Program Files\AXI System\v810\5.0\calib** and delete the following files as listed in the Table 3.

System Model	Directory	Files Required to be deleted
V810 Standard	C:\Program Files\AXI System\v810\5.0\calib	i. throughput.hardware.calib.binary ii. throughput.psp.calib.binary
	C:\Program Files\AXI System\v810\5.0\log\calib\OpticalCalibration	All .png images
	C:\Program Files\AXI System\v810\5.0\log\calib\OpticalSystemOffsetAdjustment	All .png images

Table 3 Previous Calibration Files

3. Open V810 GUI and run 1 loop of Confirmation, Diagnostics and Adjustment (CD&A)
Remarks: Only proceed to Step 4 if CD&A test pass.
4. Go to the following directory: C:\Program Files\AXI System\v810\5.0\config
Open **software.config** and set the **enablePsp=false** to **enablePsp=true** as shown in the Figure 3 below.

Save the software.config and close the software.config after modify the keys.

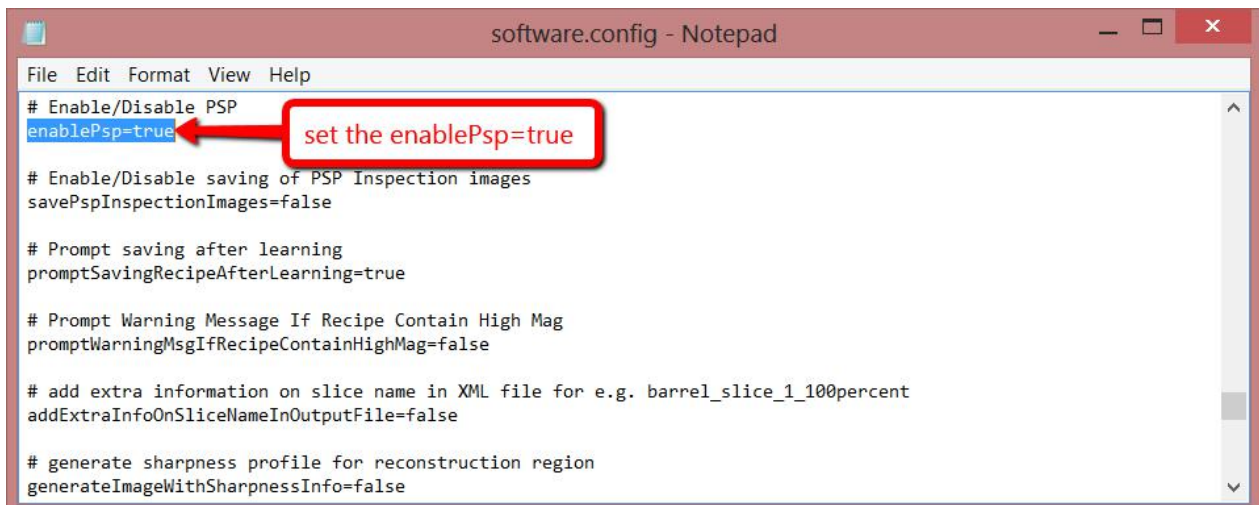


Figure 3 Software.config

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5. Go to the following directory: C:\Program Files\AXI System\v810\5.0\config

Open hardware.config and set the following keys at the hardware.config which mentioned below (refer to Figure 4):

pspModuleVersion=2

pspCalibrationVersion=4

pspSetting=1

opticalCameraModel=axiOpticalCameraType2

Save and close hardware.config after change the keys

```

#####
## PSP Module settings
##
## Version 1 = Legacy PSP module (with DLP Projector)
## Version 2 = New PSP module (with EBLE projector)
#####
pspModuleVersion=2
#####
## PSP Calibration settings
##
## Version 1 = Manual-loaded PSP calibration jig
## Version 2 = XXL On-Stage Calibration jig (old jig with 5 plane)
## Version 3 = On-Stage Calibration jig (new jig with 3 plane) - For Std and XXL
## Version 4 = New Manual-loaded PSP calibration jig (with 3 plane) - For Std Only
#####
pspCalibrationVersion=4
#####
## PSP Module Connection settings
##
## 1 = Connect to Both Front and Rear Module
## 2 = Connect to Front Module Only
## 3 = Connect to Rear Module Only
#####
pspSetting=1
#####
## PSP On-Stage Jig Object Plane Z-Height (in micron)
#####
pspOnStageJigObjectPlaneZHeightInMicron=4300
#####
## Optical camera settings
#####
opticalCameraModel=axiOpticalCameraType2
opticalCameraLoggingEnabled=false
#####
## Micro-Controller settings
#####
pspMicroControllerSerialPortName=COM4
#####
## Ethernet-Based Light Engine
##
## Supported Setting
## 1) Current (mA) : 350 (default), 495, 635, 770
## 2) Period : 20 (default), 30, 40, 50
#####
ethernetBasedLightEngineHostIpAddress=123.123.123.200
ethernetBasedLightEngineCurrentSettingMicroAmps=350
ethernetBasedLightEnginePeriod=20

```

Figure 4 Hardware.config

6. Open V810 GUI and run 1 loop of Confirmation, Diagnostics and Adjustment (CD&A)

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7. Go to the Options -> Production Options -> General and change the following items as shown in the Figure 5.

Loading mode: Pass Back

Flow direction: Left To Right

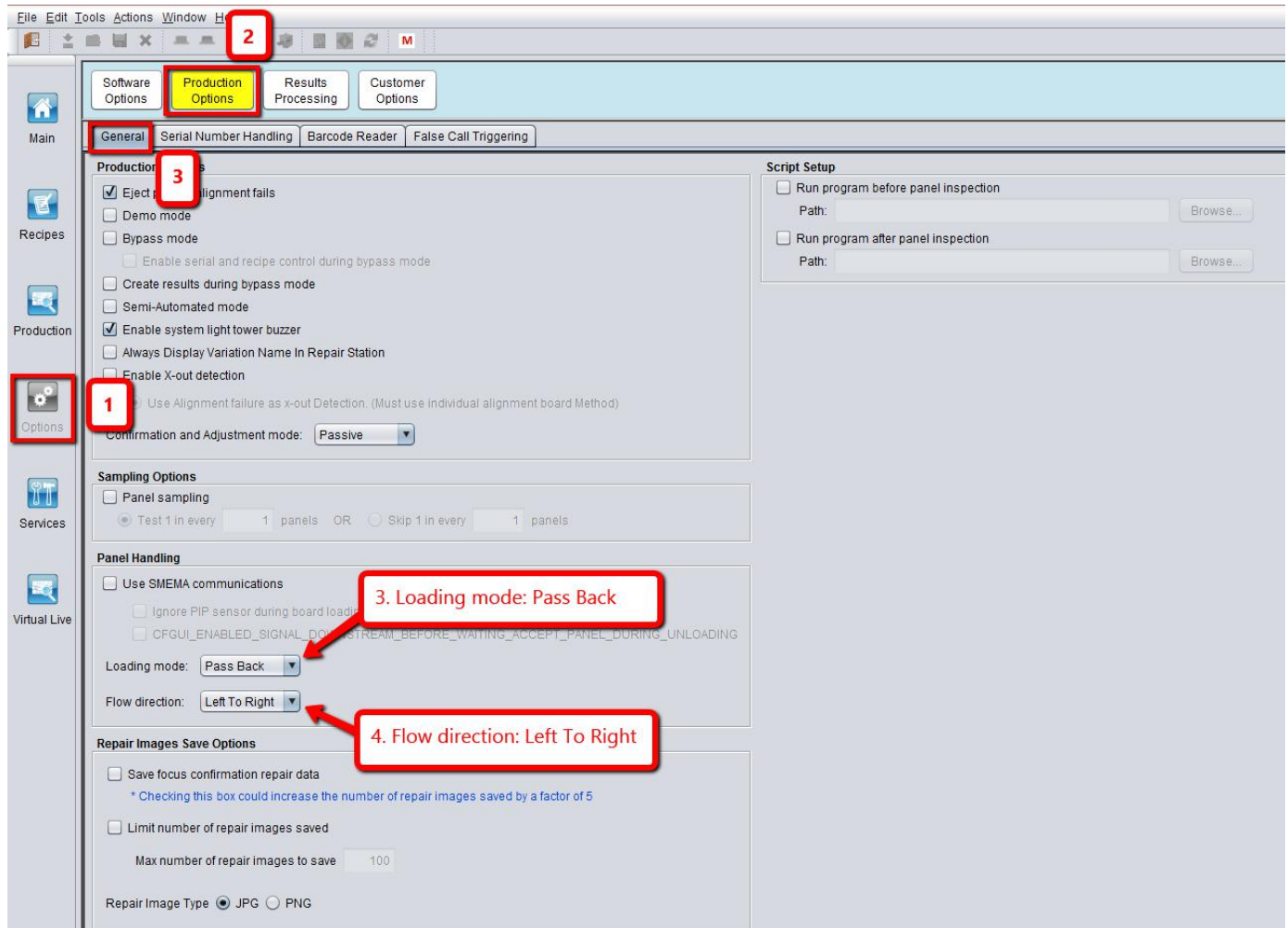


Figure 5 Loading Mode

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8. At Confirmation, Diagnostics and Adjustment (CD&A), under the Manual Optical Adjustments, right click **Optical System Fiducial Adjustment** and select **Run test without dependent tests** as shown in the Figure 6.

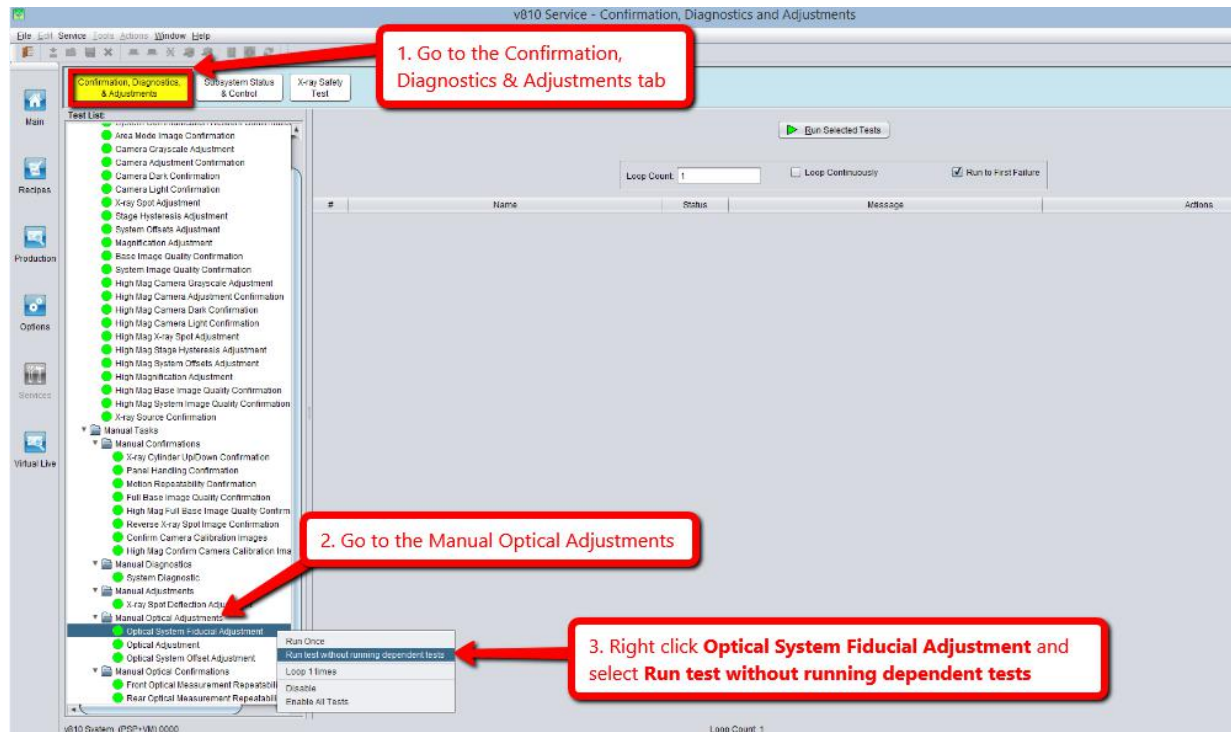


Figure 6 Optical System Fiducial Adjustment

9. Once the Adjusting optical system fiducial reach 10% complete as shown in the Figure 7, load the manual load adjustment plate. May refer to Figure 7 to Figure 9.

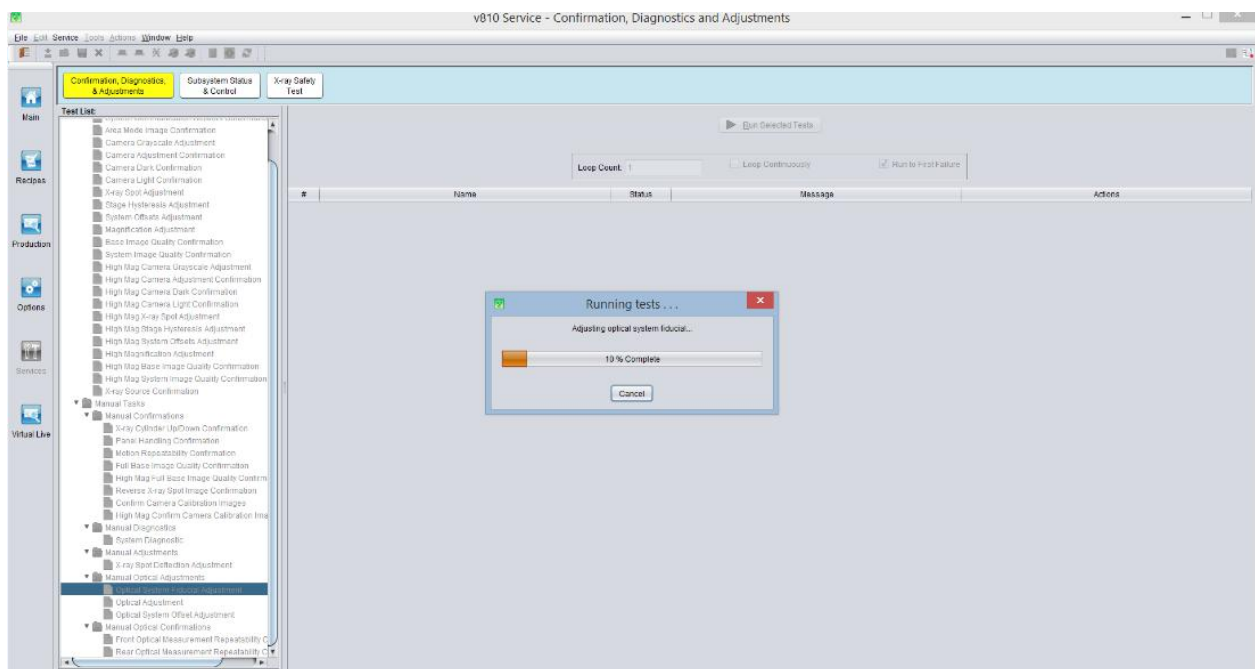


Figure 7 Loading the adjustment plate to the machine

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Figure 8 Adjustment Plate



Figure 9 Left Outer Barrier (Manual Load Jig)

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10. Once the Adjustment Plate has been loaded, the image from the Front PSP Optical Camera will be shown as in the Figure 10.

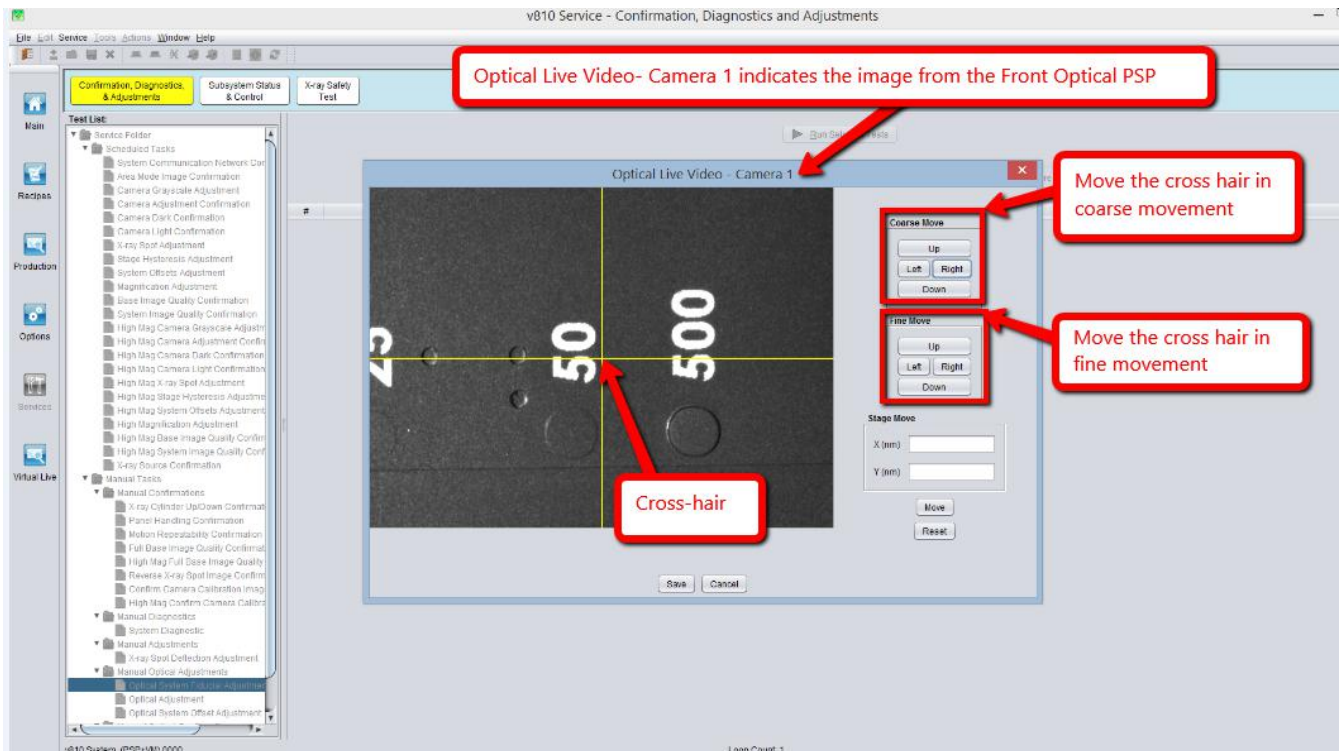


Figure 10 Front PSP Optical Camera

11. Move the cross hair to the location shown in the Figure 11. Once the location has been confirmed, click save as shown in the Figure 11.

Remarks: Location of the cross hair require to be within middle of the circle as shown in Figure 12.

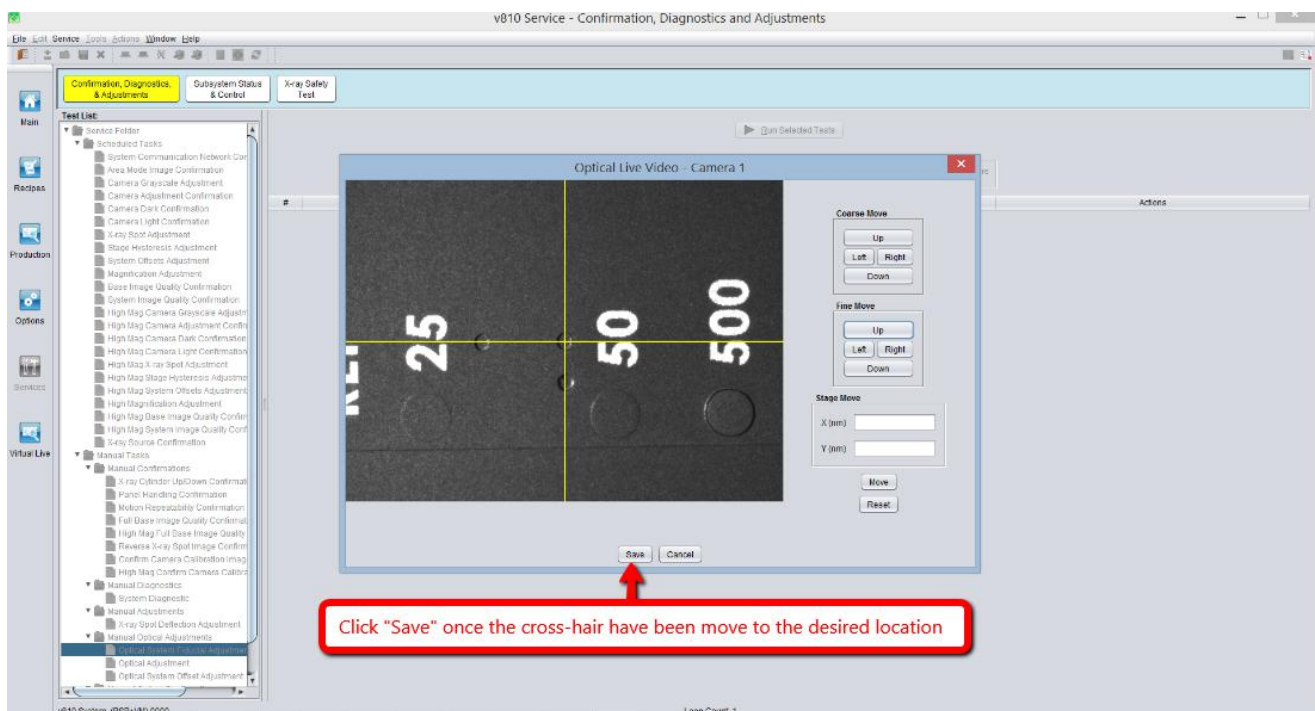


Figure 11 Front PSP Module Adjustment

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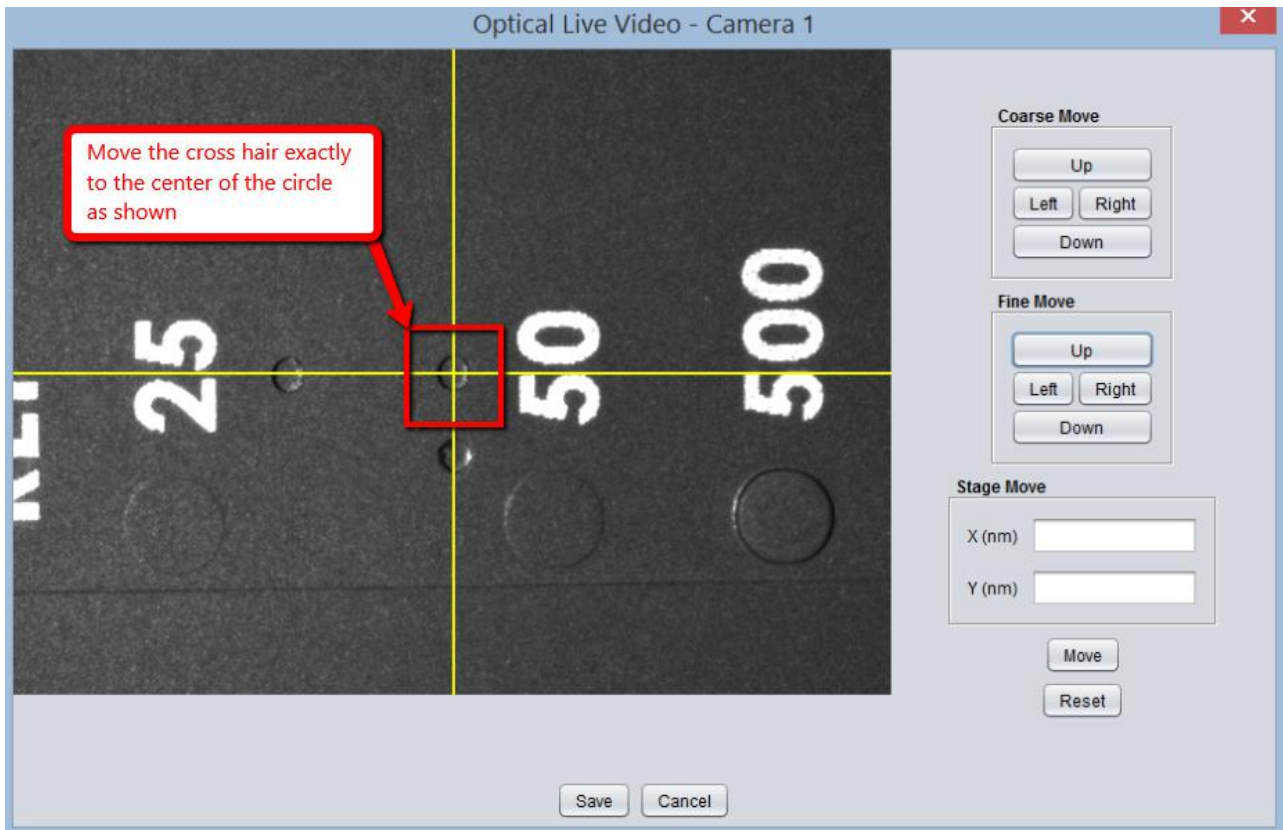


Figure 12 Exact point to select the cross hair

12. After performing Step 11, the image from the Rear PSP Camera will be shown as in Figure 13.

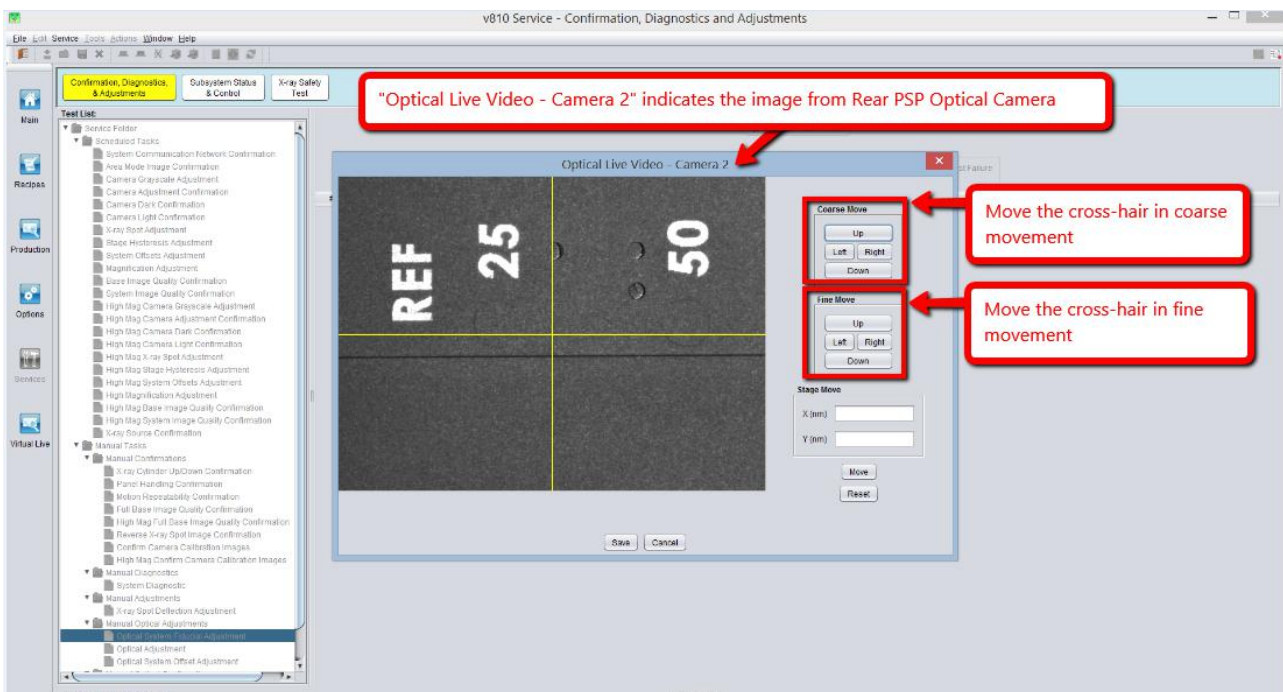


Figure 13 Rear PSP Optical Camera

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13. Move the cross hair to the location shown in the Figure 14. Once the location has been confirmed, click save as shown in the Figure 14.

Remarks: Location of the cross hair require to be within the middle of the circle as shown in Figure 15.

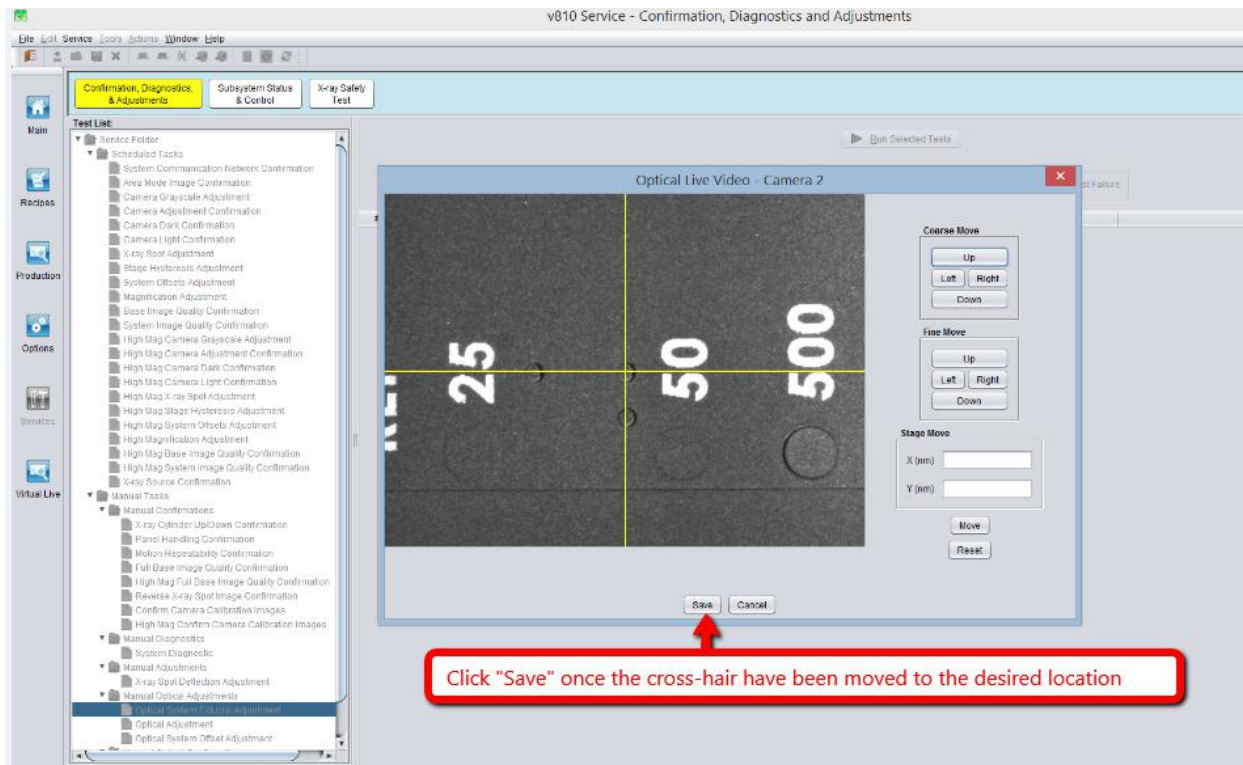


Figure 14 Rear PSP Fiducial Point choosing

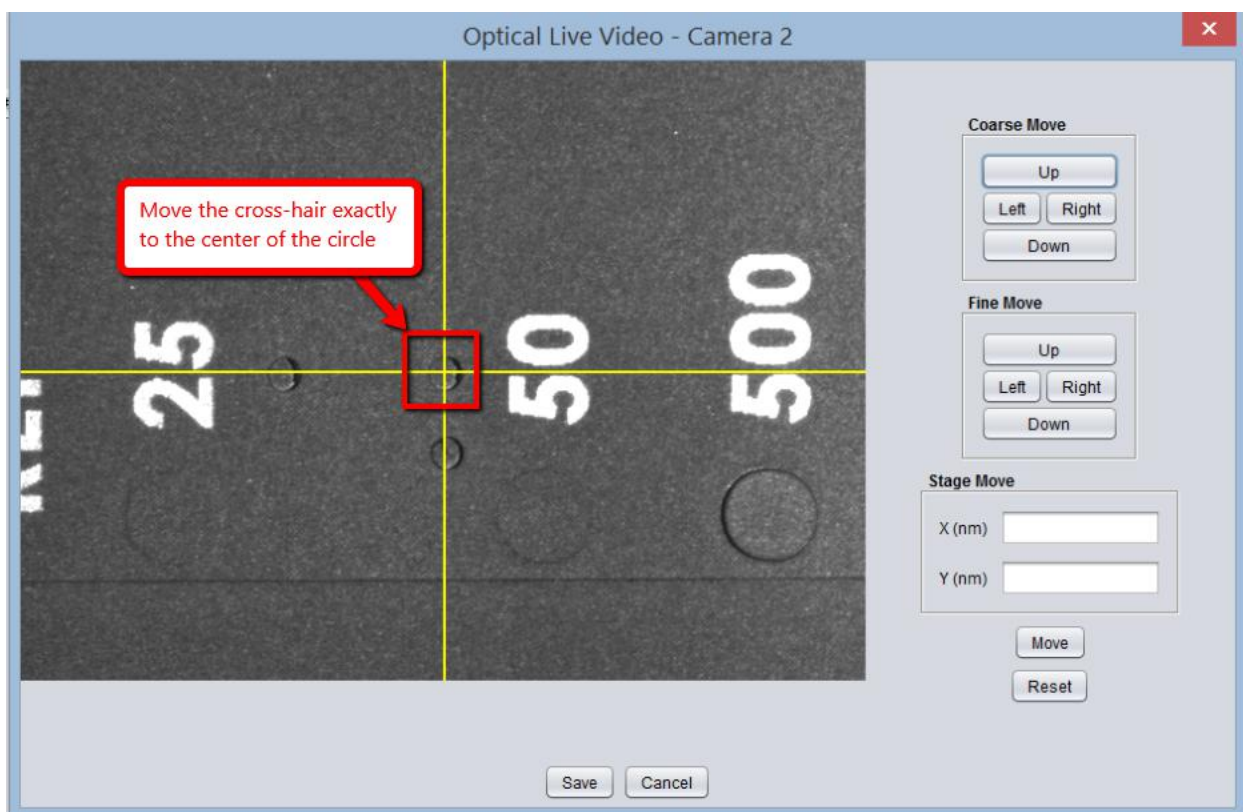


Figure 15 Exact point for cross hair selection

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14. The V810 machine will open the left outer barrier; remove the manual load jig from the machine as shown in the Figure 16.



Figure 16 Remove the JIG

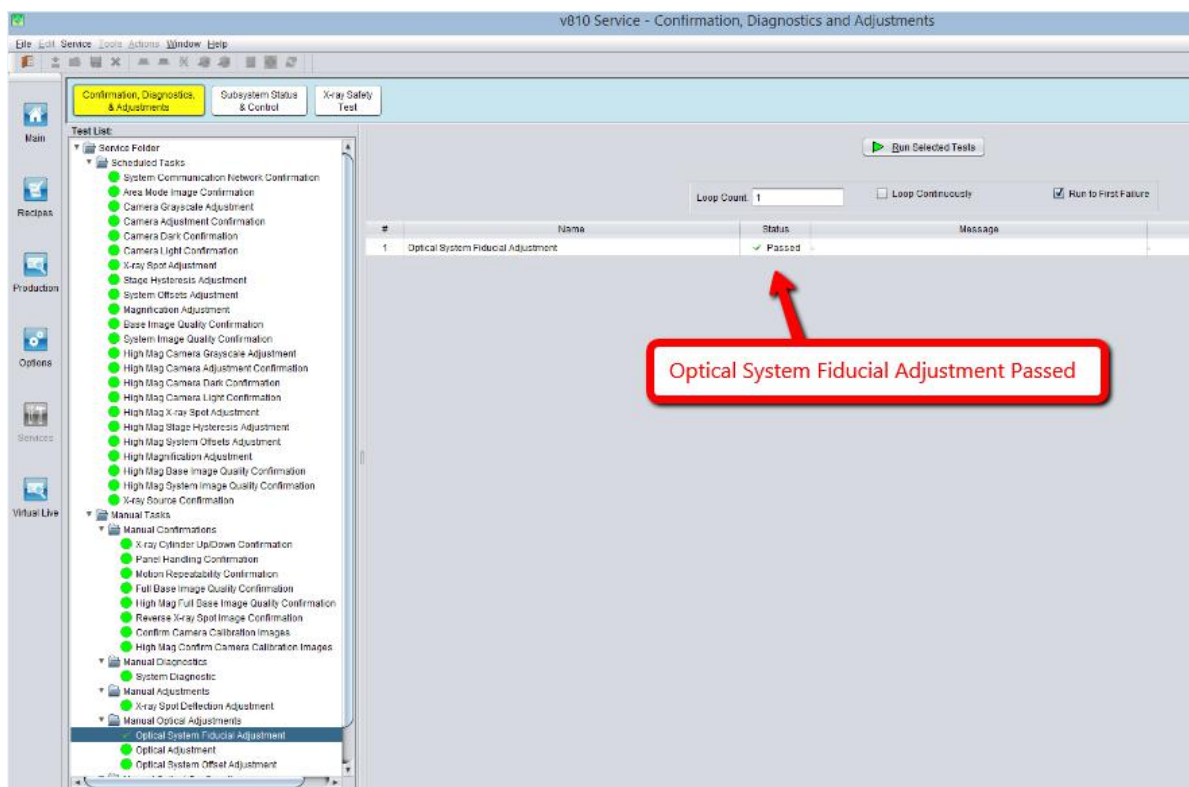


Figure 17 Optical System Fiducial Adjustment Passed

15. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Adjustments**, right click **Optical Adjustment** and select **Run test without dependent tests** as shown in the Figure 18.

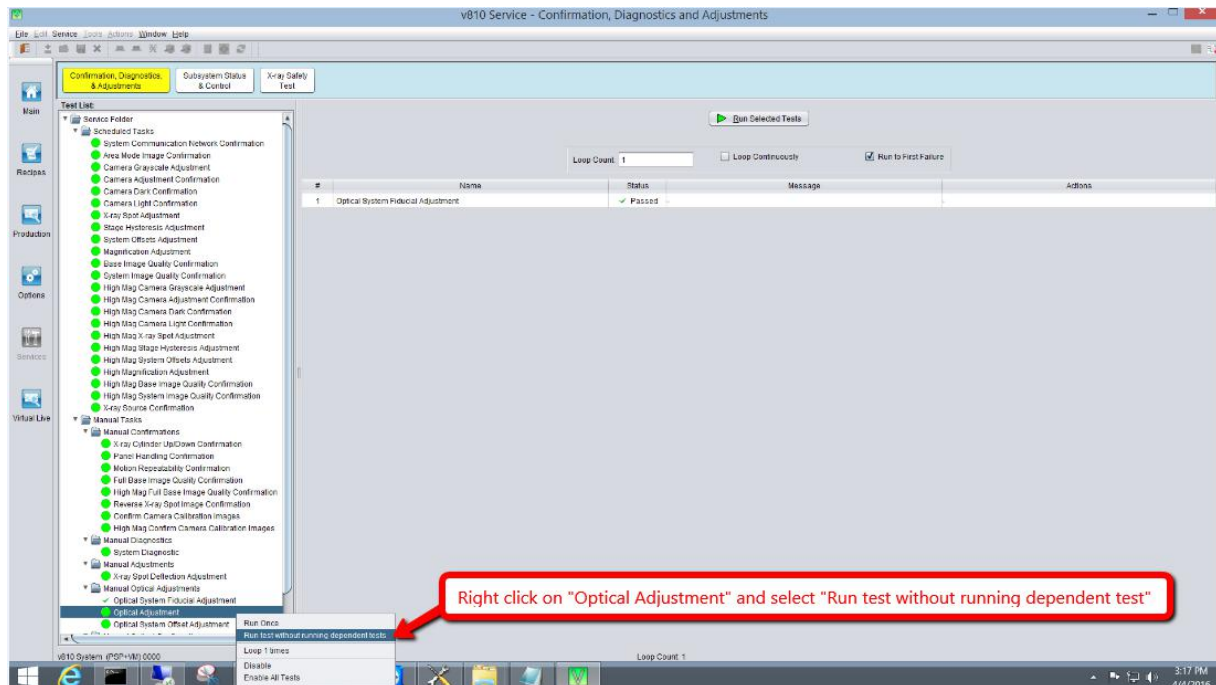


Figure 18 Optical Adjustment

16. Insert the jig once the left outer barrier open as shown in the Figure 19. Once the test is finished, the outer barrier will open. Remove the jig as shown in the Figure 20.



Figure 19 Jig loading

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Figure 20 Jig removal

17. Once the jig is removed as shown in the Step 16 and the outer barrier is closed, the status of the test will be shown in the Figure 21.

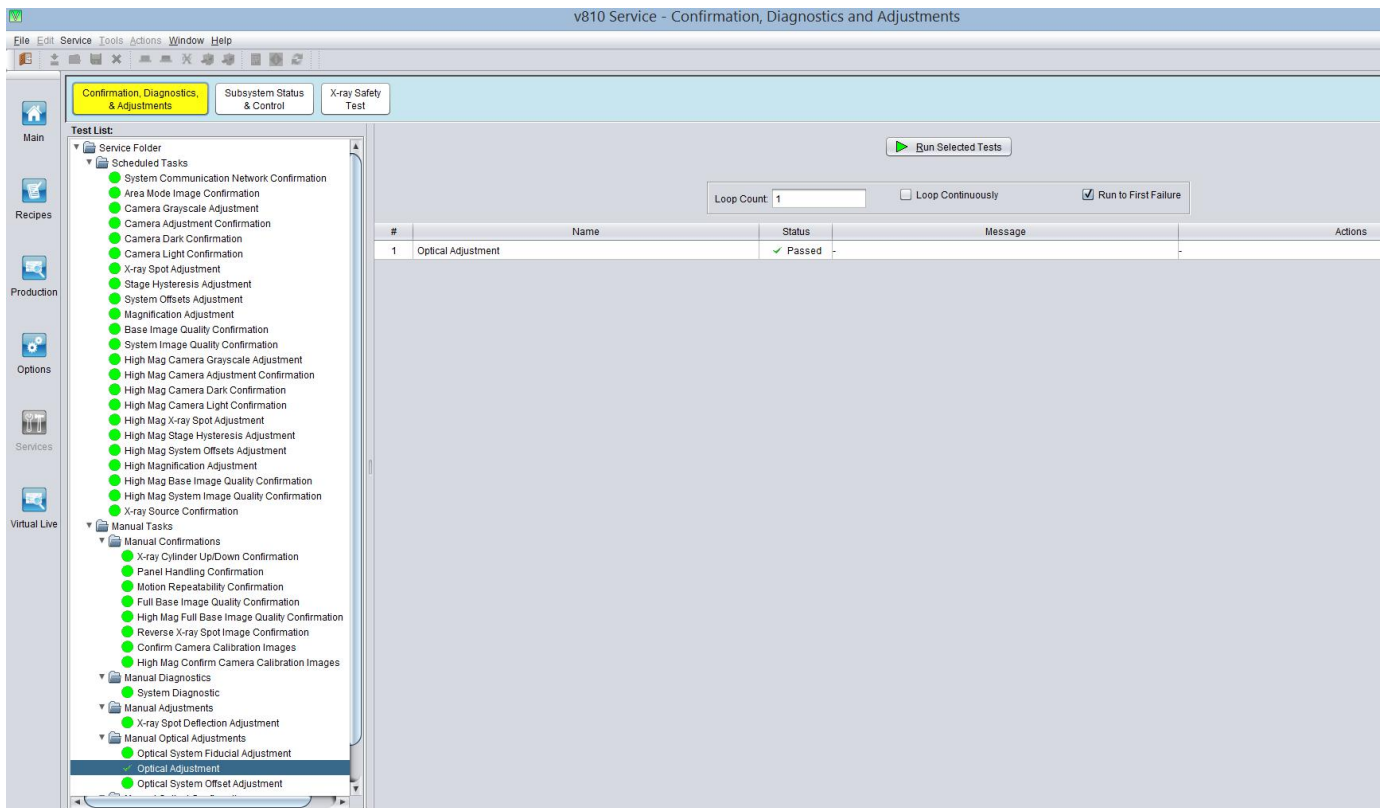


Figure 21 Optical Adjustment

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18. Go to the following directory

Directory: C:\Program Files\AXI System\v810\5.0\log\calib\OpticalCalibration

Total number of 32 images will be generated as shown in the Figure 22.

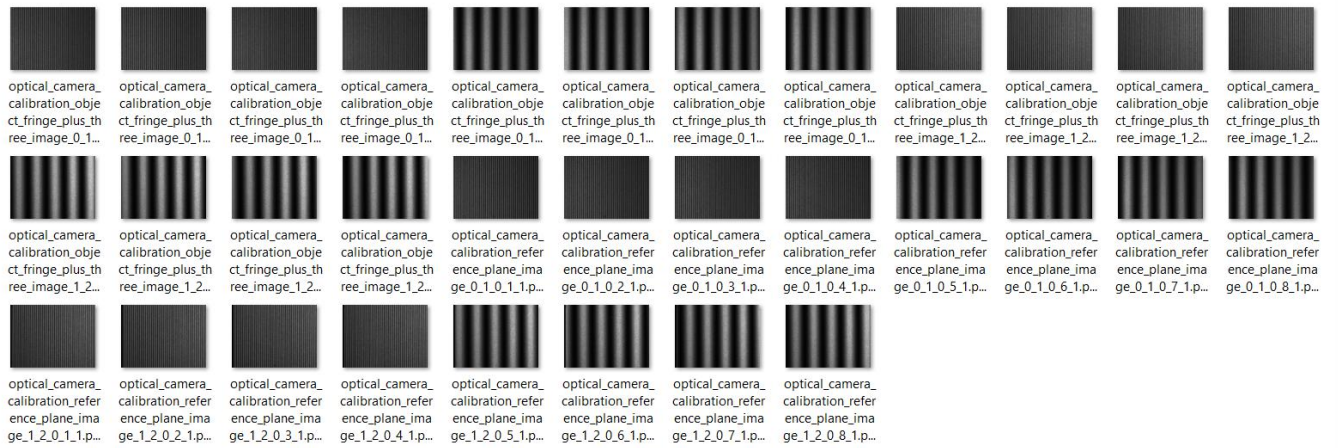
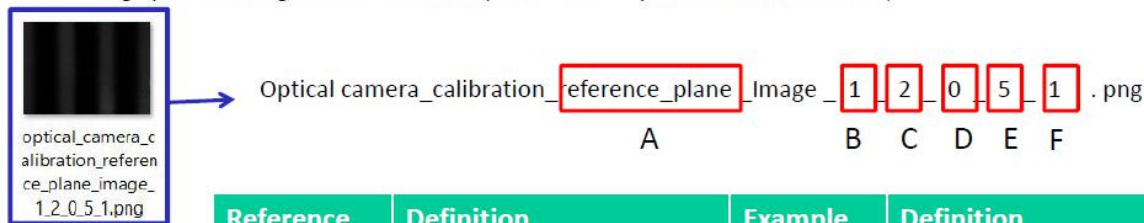


Figure 22 Optical Calibration Fringe Pattern

Additional reference for the Fringe Pattern Naming is as in Figure 23.

Fringe pattern image name definition (Store under Optical calibration folder)



Reference	Definition	Example	Definition
A	Calibration image captured by using REF plane at PSP calibration jig	Reference_Plane	Calibration image captured by using REF plane at PSP calibration jig
B	Front Camera (id = 0) Rear Camera (id = 1)	1	Rear camera
C	The value will always 1 value higher than Camera Id. The values are 1 (front camera) and 2 (rear camera)	2	The value will always 1 value higher than Camera Id. Means If "B" is indicate "1", this device ID will add on "1" at "B", so the camera device ID will be 2
D	Profile 1 (id = 0), Profile 2 (id = 1)	0	Default will set to "0" Means just using "REF" & "+3" plane at PSP jig only
E	For PSP 2.0, the sequence number starts from 1 to 8.	5	Capturing 5th of fringe pattern Sequence number starts from 1 to 8.
F	This is designed so that we can handle multiple run of calibration for averaging purpose. For now, it is default to 1.	1	Default set to "1", means just capturing one image at one time only

Figure 23 Fringe pattern naming

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19. Refer to the Figure 24 and Figure 26 for the sample of the good adjustment images and Figure 25 and Figure 27 for the sample of bad adjustment images.

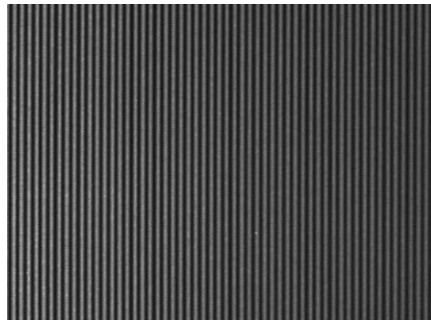


Figure 24 Good Images

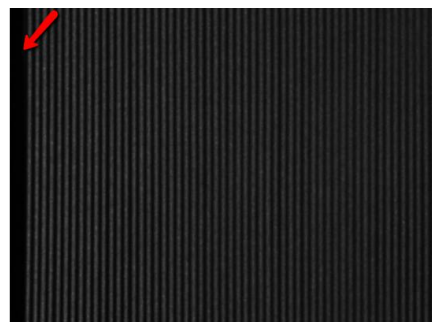
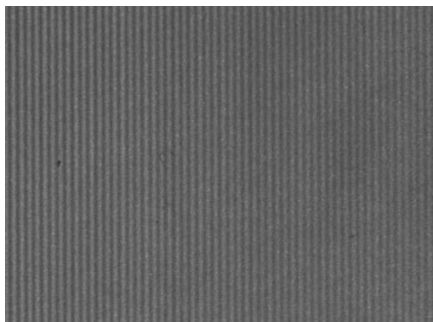


Figure 25 Bad Images

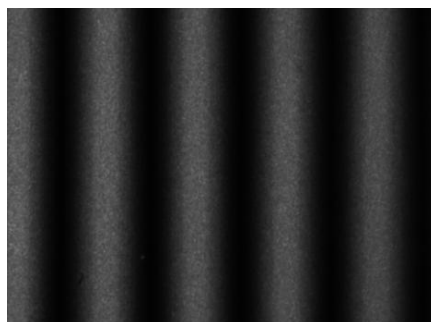


Figure 26 Good Images

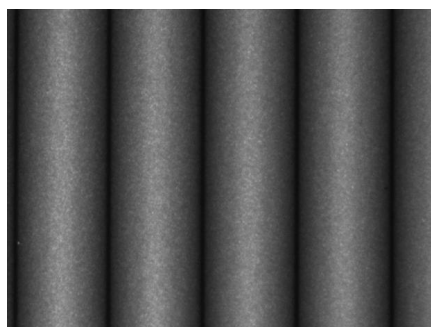


Figure 27 Bad Images

If adjustment is require,may refer to the procedure“AXI V810 Front Module Offline Adjustment Procedure (FOR FSE) ” and “AXI V810 PSP Rear Module Offline Adjustment Procedure (FOR FSE) ”

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20. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Adjustments**, right click **Optical System Offset Adjustment** and select **Run test without dependent tests** as shown in the Figure 28.

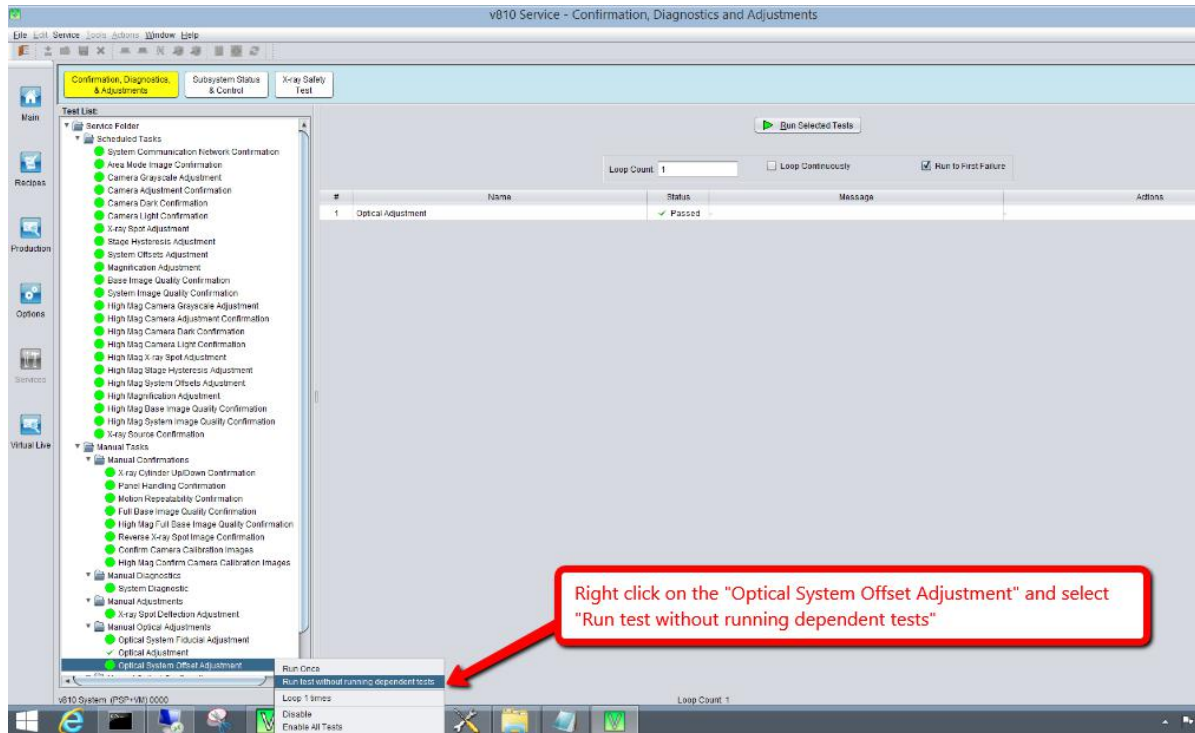


Figure 28 Optical System Offset Adjustment

21. Insert the jig once the left outer barrier open as shown in the Figure 29. Once the test is finished, the outer barrier will open. Remove the jig as shown in the Figure 30.

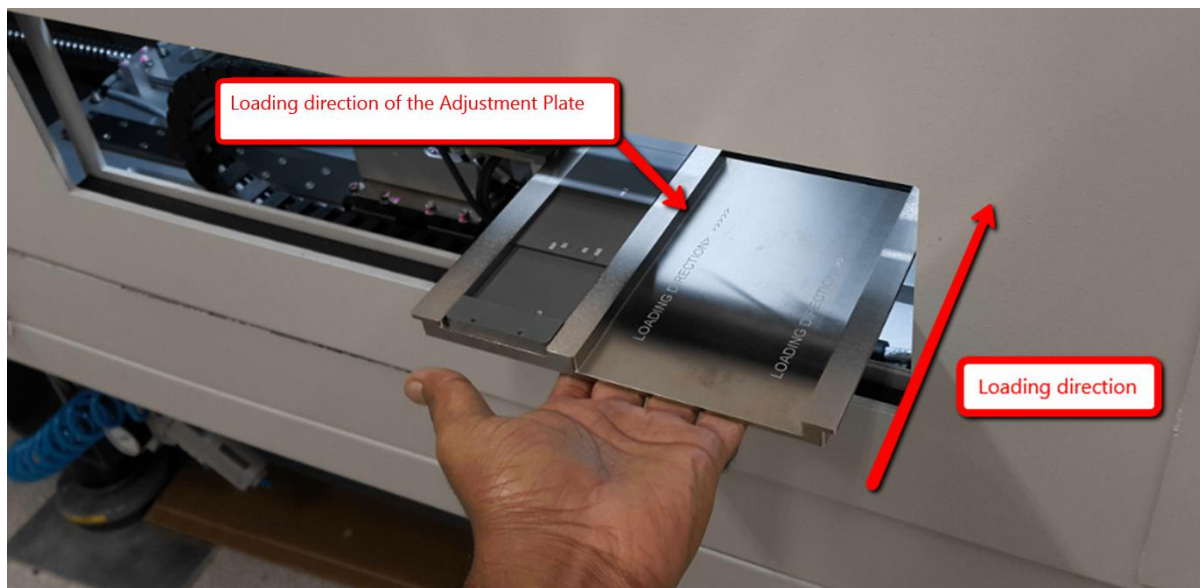


Figure 29 Jig loading

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Figure 30 Jig removal

22. Once the jig is removed as shown in the Step 18, the status of the test will be shown as in Figure 31.

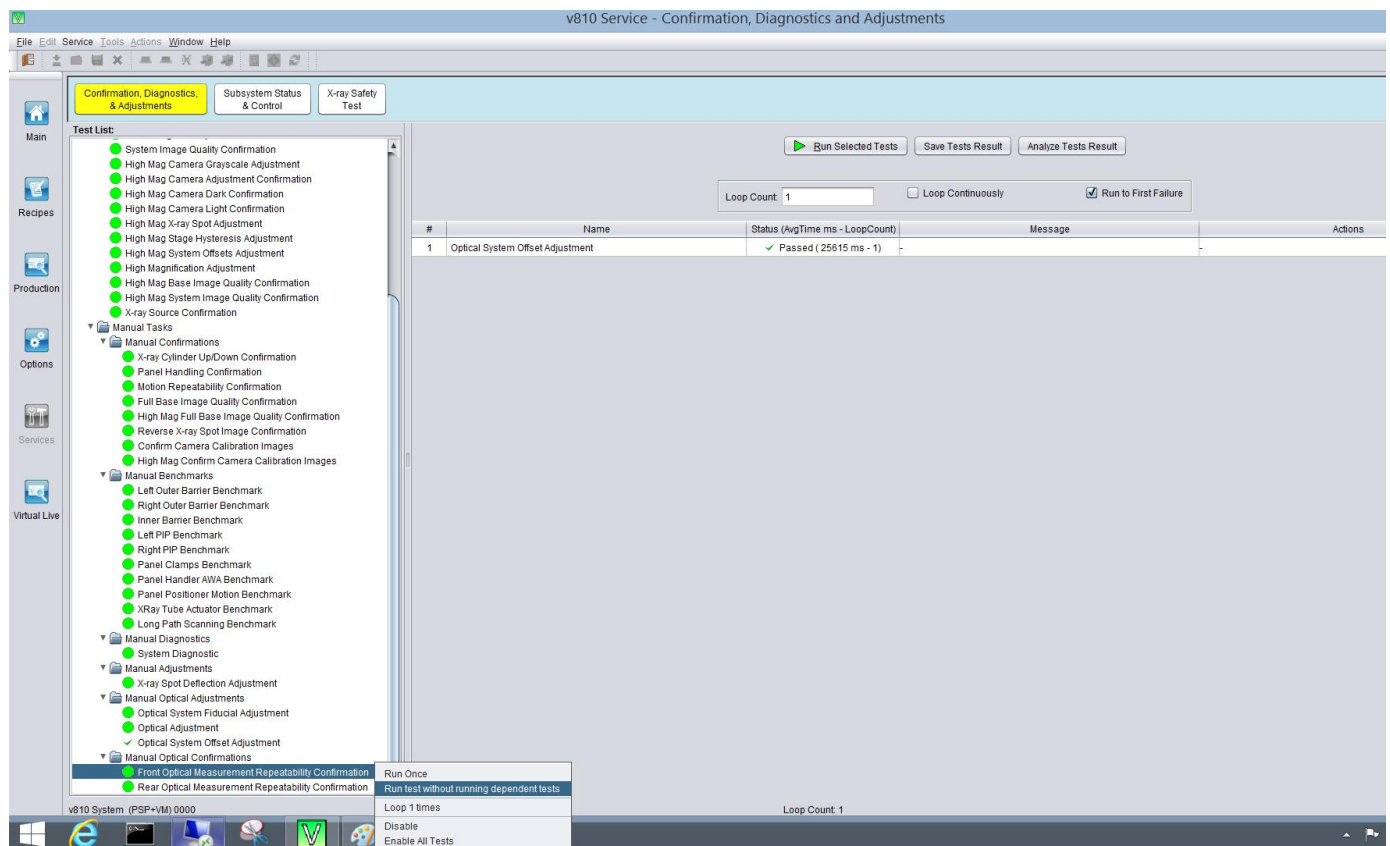


Figure 31 Optical System Offset Adjustment

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23. Once the test in Step 22 is complete, go to the following folder.

Directory: C:\Program Files\AXI System\v810\5.0\log\calib\OpticalSystemOffsetAdjustment

Check the latest text.log files as shown in the Figure 32.

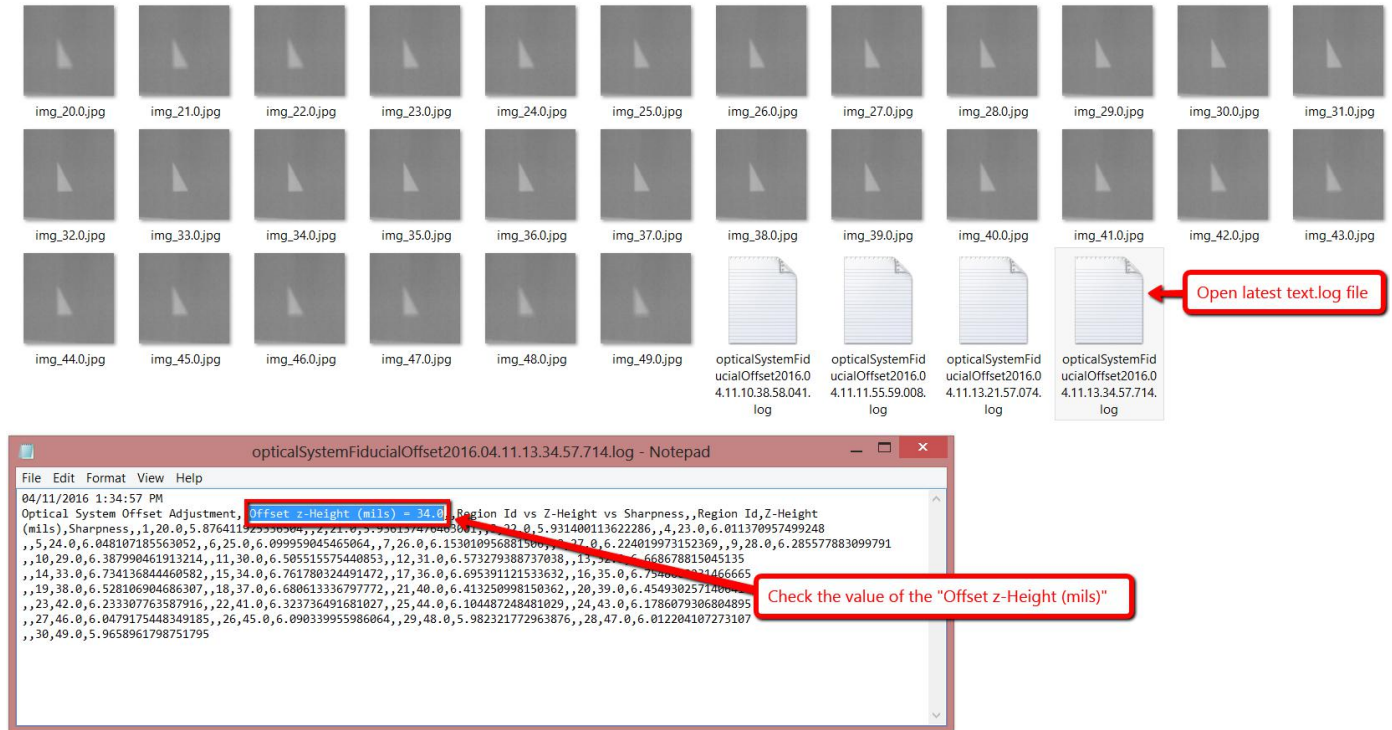


Figure 32 Z-Offset

Check the value of the Offset z-Height (mils) whether it is within the range listed in Table 4.

System Type	Low Limit	High Limit
Standard	-95 mils	- 75 mils

Table 4 Z-Offset Limit

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24. If the Optical System Offset Adjustment failed as shown in Figure 33, go to the following folder
Directory: *C:\Program Files\AXI System\v810\5.0\log\calib\OpticalSystemOffsetAdjustment*
Check the images generated inside the folder whether it is similar to images as in Figure 34 or Figure 35.

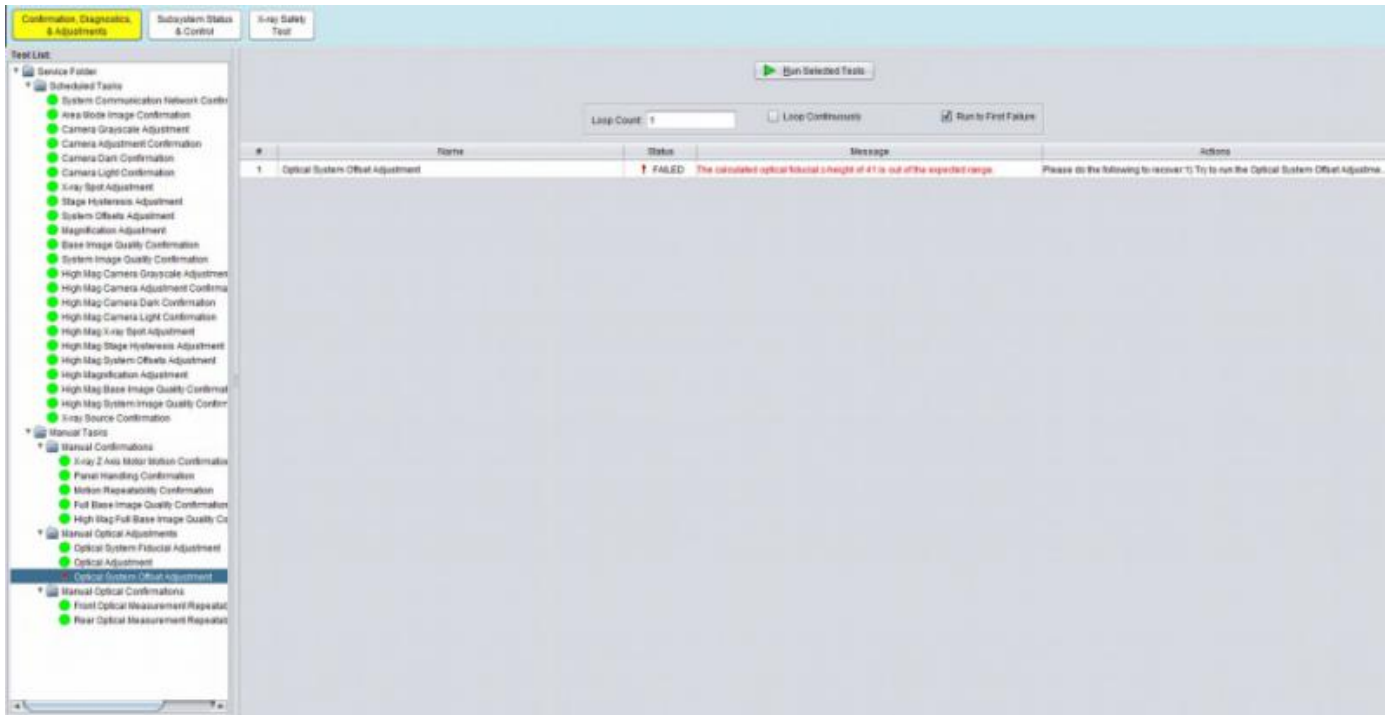


Figure 33 Optical System Offset Adjustment

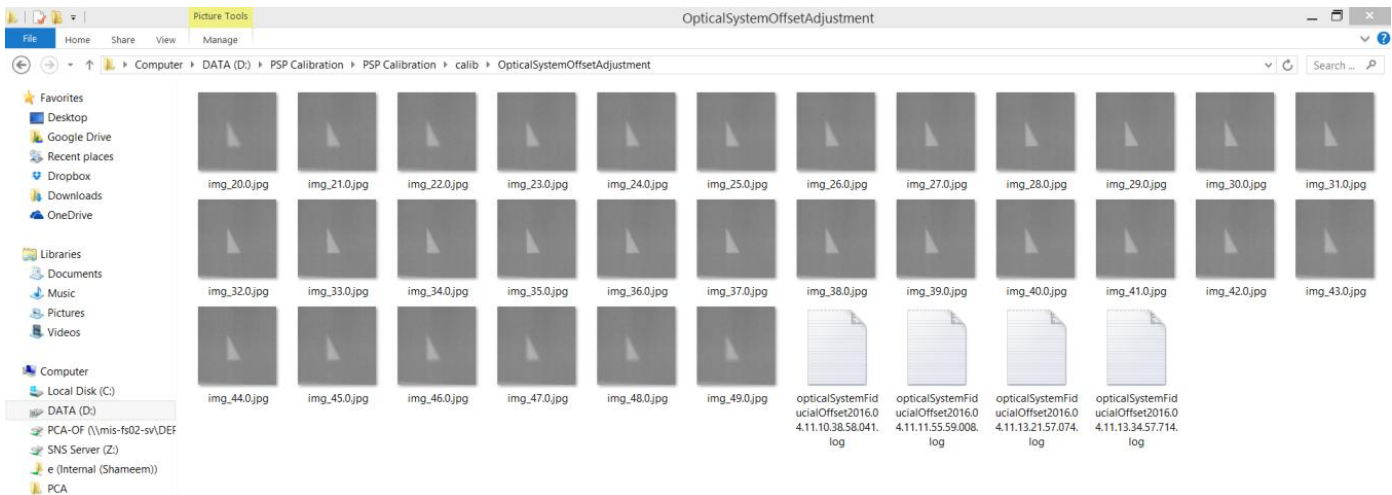


Figure 34 Good Images

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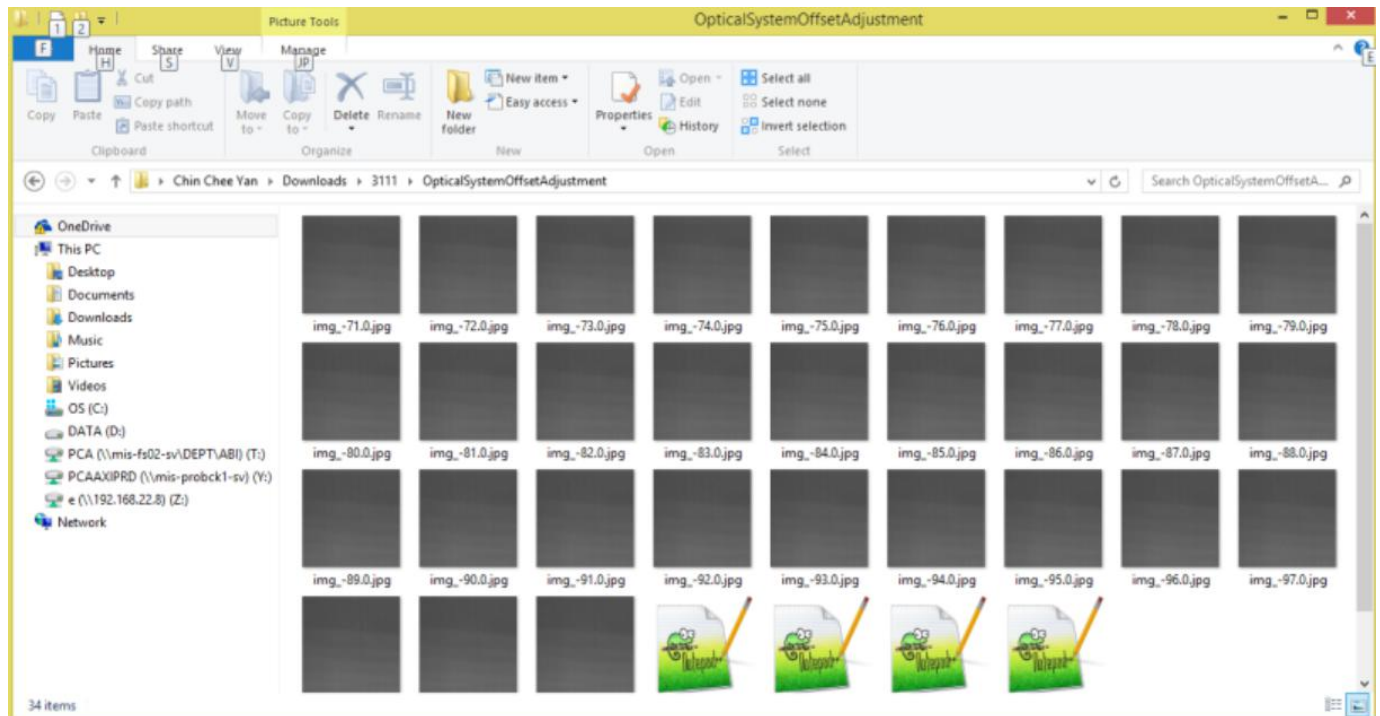


Figure 35 Bad Images

If the images generated are as in Figure 35,

- i. Require to check the existence of Fiducial Coupon which installed at bottom part of the manual load jig



Figure 36 Fiducial Coupon

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25. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Confirmation**, right click **Front Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 37.

Remarks: The first run will mainly for 500 micron test.

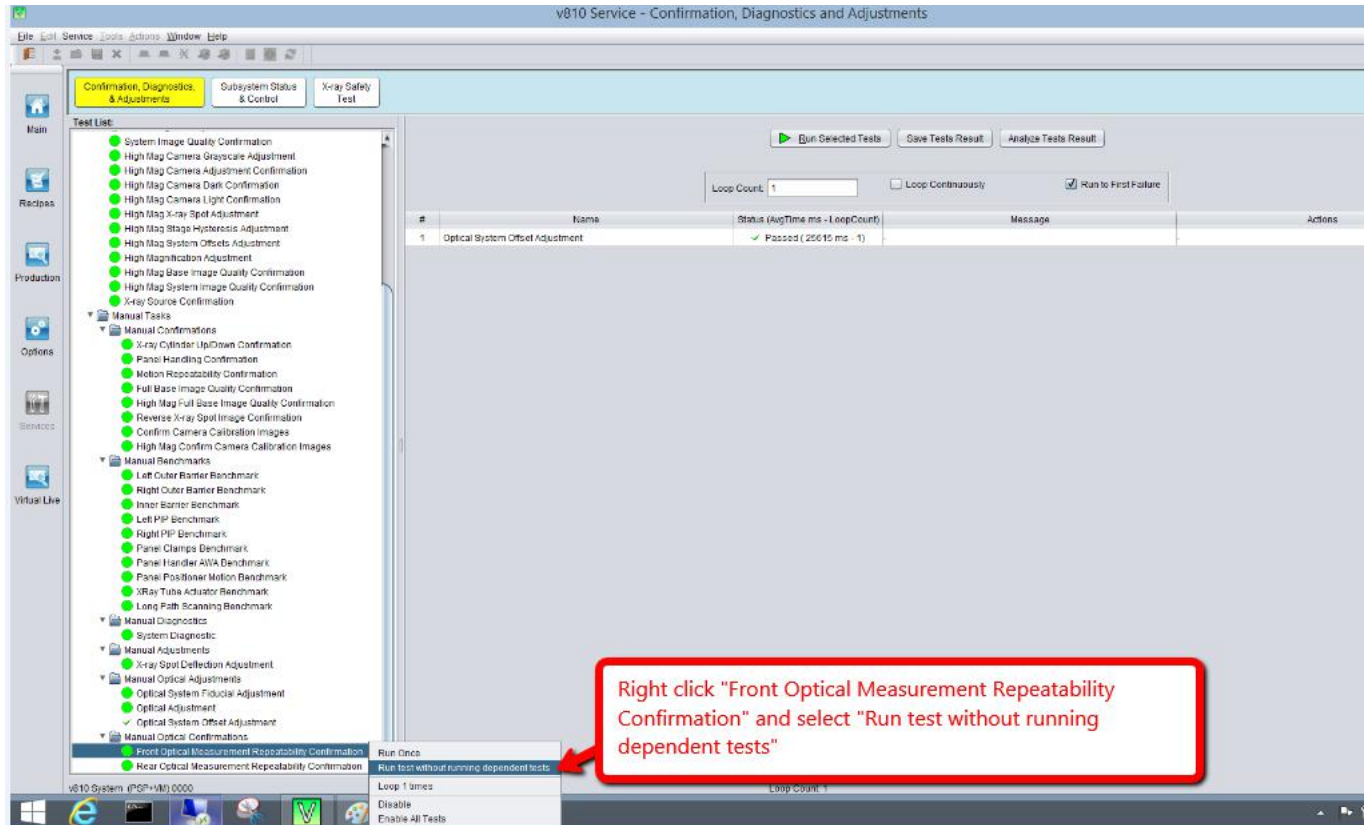


Figure 37 Front Optical Measurement Repeatability Confirmation

26. Insert the jig once the left outer barrier open as shown in the Figure 26.

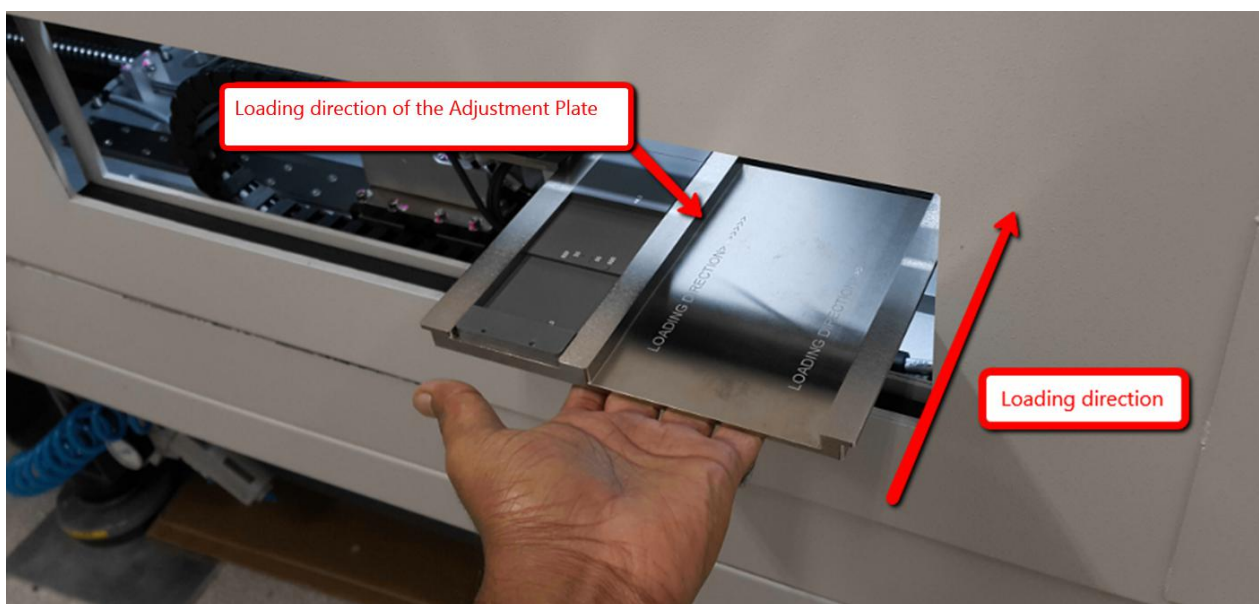


Figure 38 Jig loading

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27. Select the points of the adjustment dot as shown in the Figure 39 before proceed with the Step 28.

Remarks: Click to choose the point.

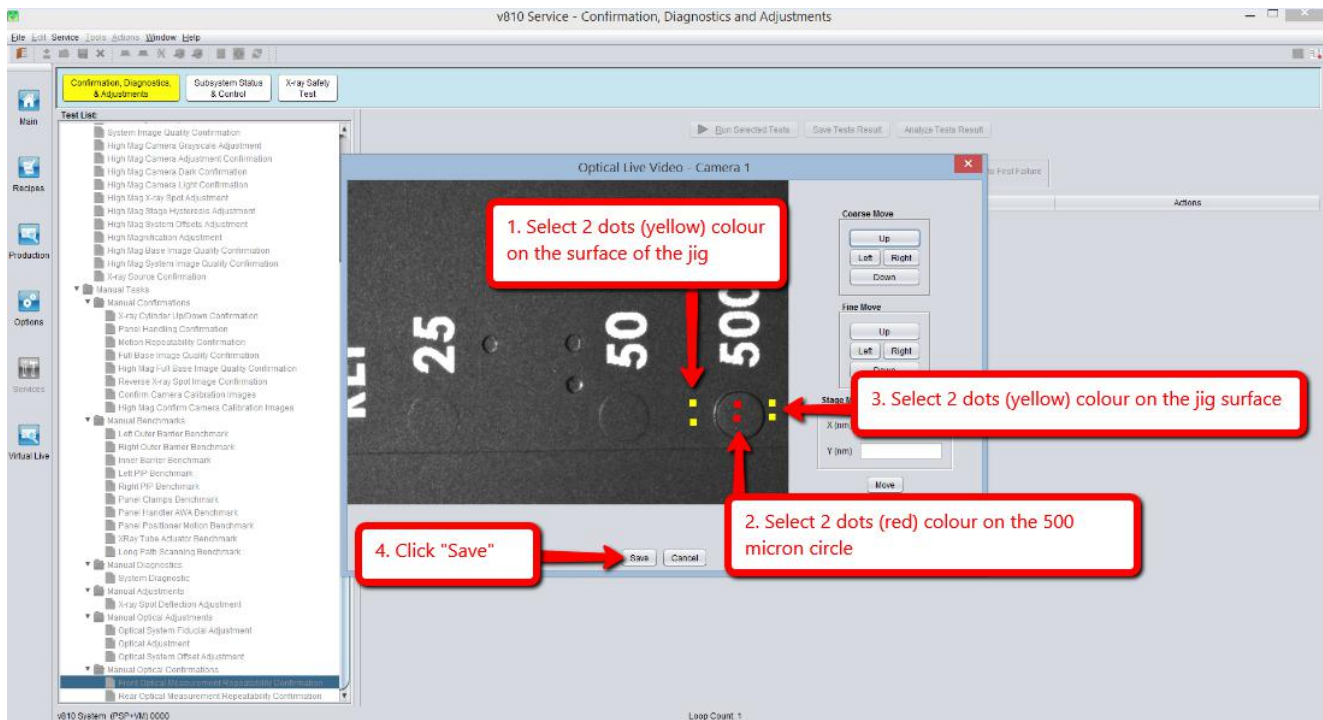


Figure 39 Adjustment Points

28. Once the outer barrier open as shown in the Figure 40, remove the on rail jig from the machine.

After the outer barrier close, the GUI will show the test have passed or failed as shown in the Figure

41.



Figure 40 Jig Removal

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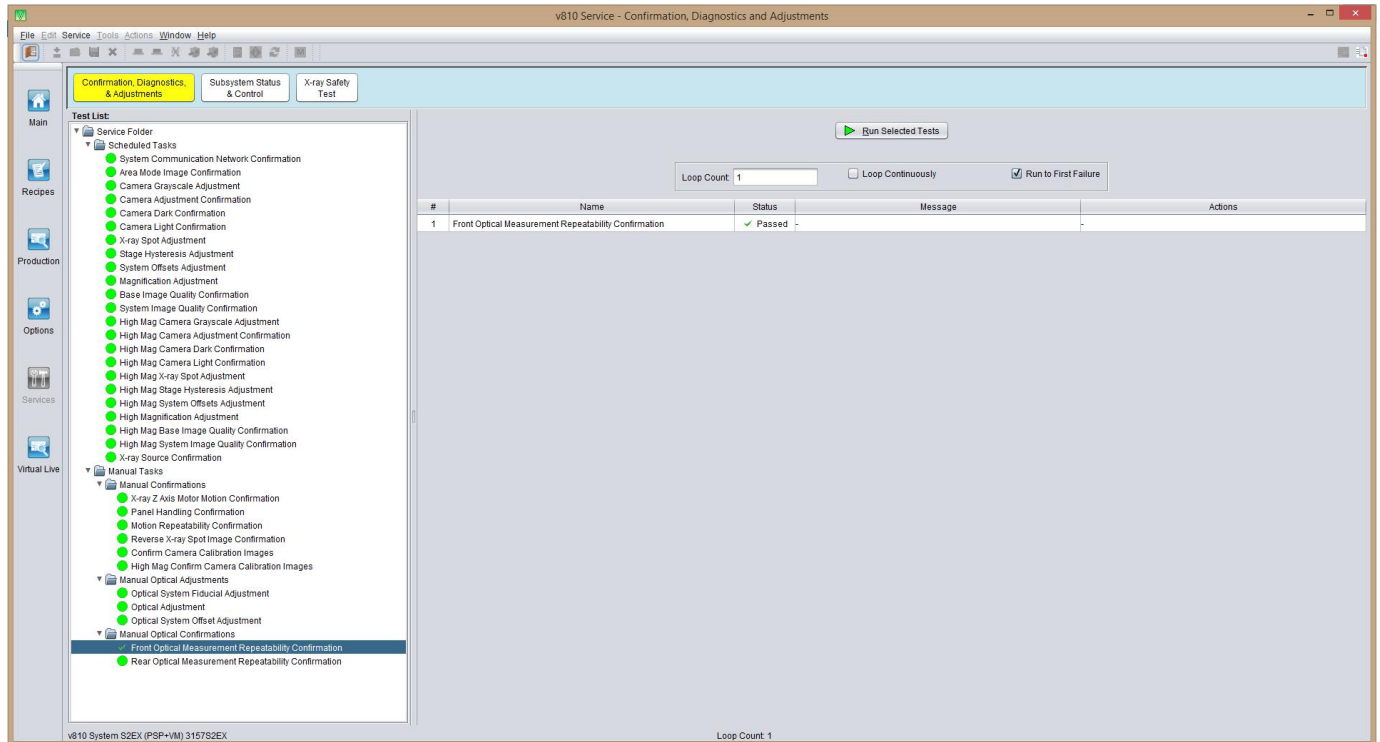


Figure 41 Front Optical Measurement Repeatability Confirmation Passed

29. Go to the following directory

Directory: C:\Program Files\AXI

System\v810\5.0\log\confirm\FrontOpticalMeasurementRepeatability

Open the latest text.log file as shown in the Figure 42 and check the value of the “Mean Z-Height (nm)” and “StdDev Z-Height (nm)” whether those values are within the range listed in Table 5.

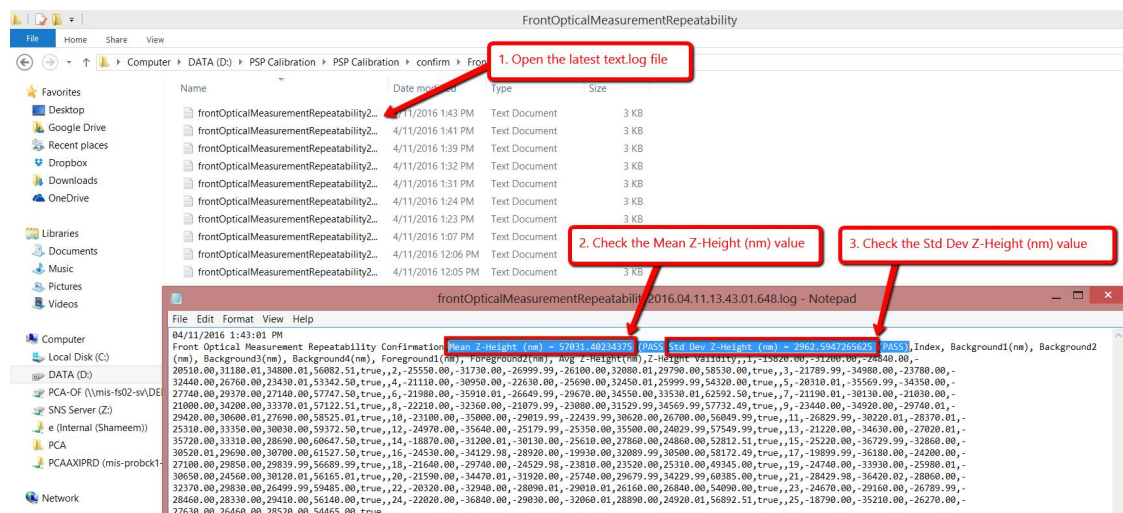


Figure 42 Repeatability test

Log files	Low Limit	High Limit
Mean Z-Height (nm)	400000.0	600000.0
StdDev Z-Height (nm)	0.0	5000.0

Table 5 500 Micron repeatability test

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30. At **Confirmation, Diagnostics and Adjustment (CD&A)** , under the **Manual Optical Confirmation**, right click **Front Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 43 below.

Remarks: The second will run for 50 micron.

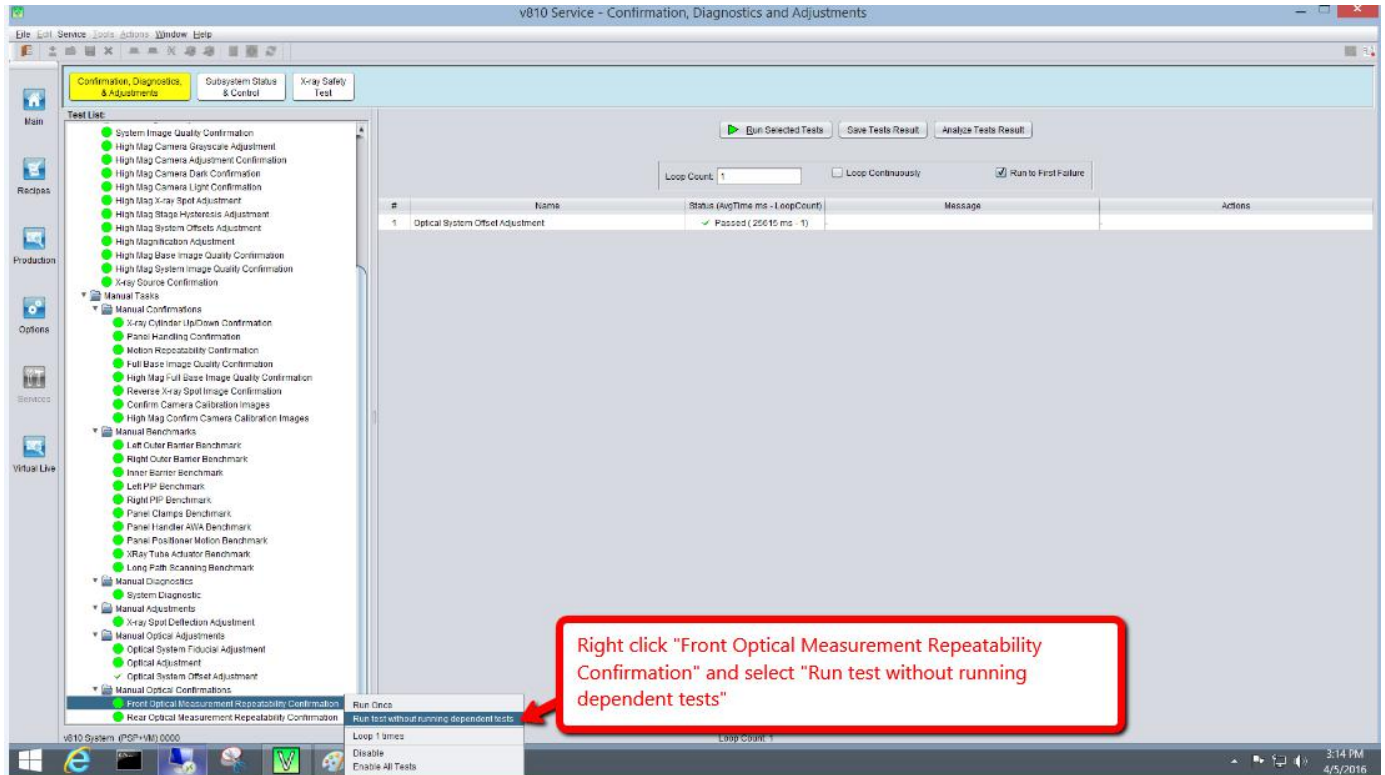


Figure 43 Repeatability test 50 micron

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31. Insert the jig once the left outer barrier open as shown in the Figure 44.

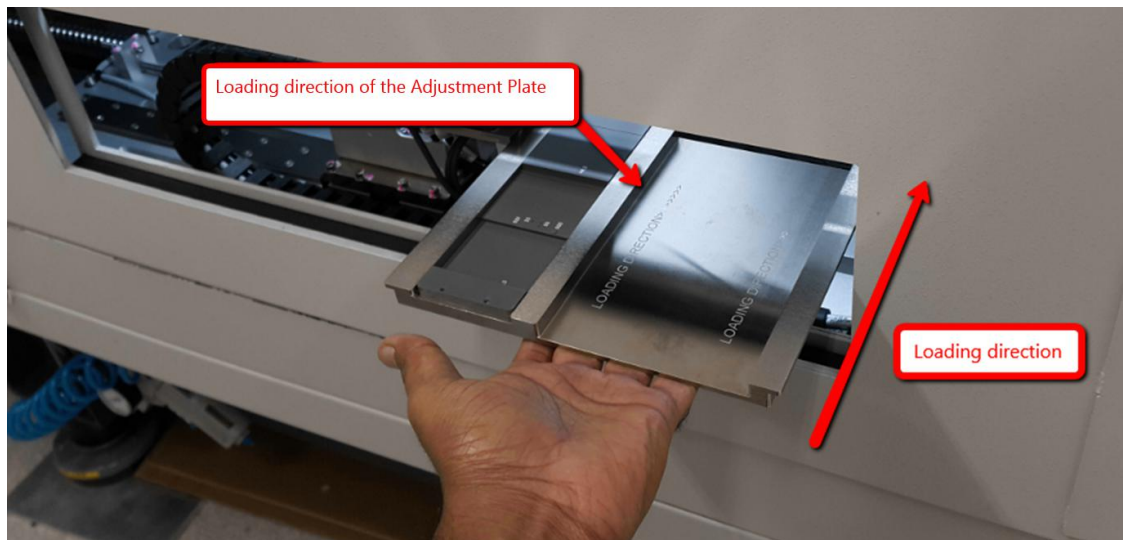


Figure 44 Jig loading

32. Select the points of the adjustment dot as shown in the Figure 45 before proceed with the Step 33

Remarks: Click to choose the point.

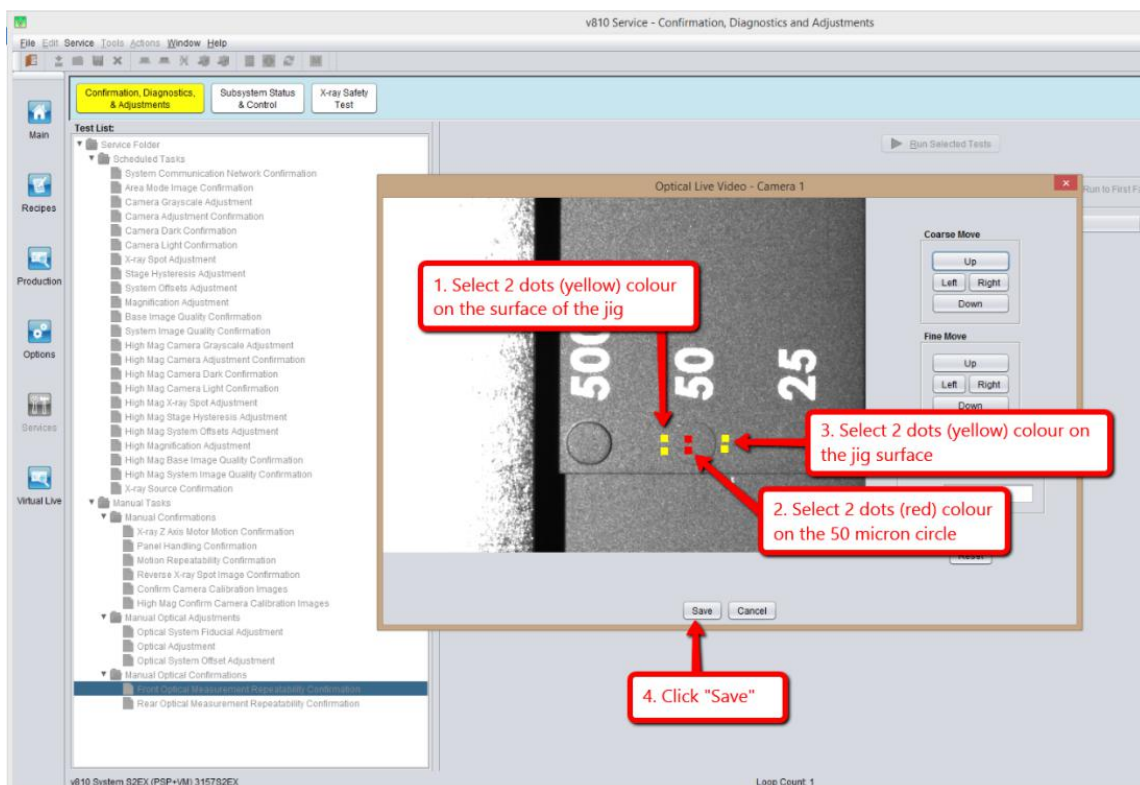


Figure 45 Adjustment Point

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33. Once the outer barrier open as shown in the Figure 46, remove the manual load jig from the machine. After the outer barrier close, the GUI will show the test have passed or failed as shown in the Figure 47.



Figure 46 Jig Removal

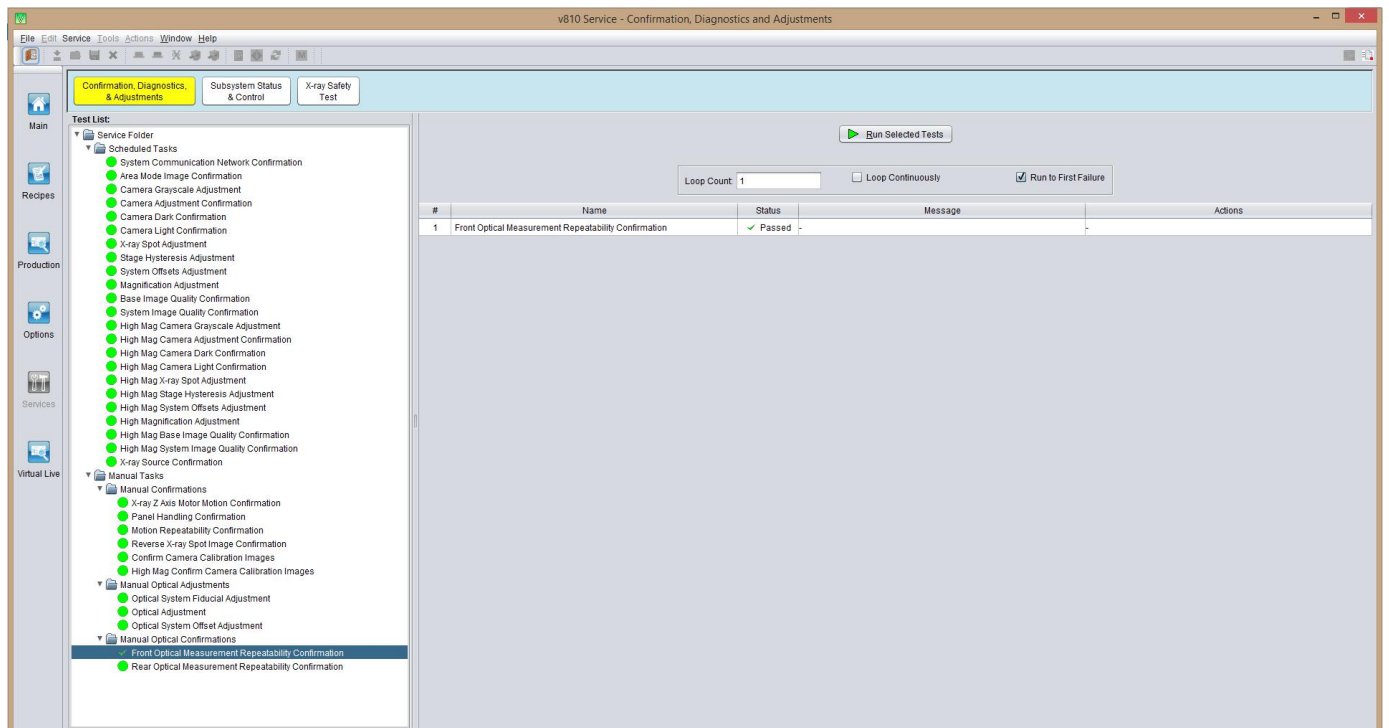


Figure 47 Front Optical Measurement Repeatability Confirmation Passed

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34. Go to the following directory

Directory: C:\Program Files\AXI

System\v810\5.0\log\confirm\FrontOpticalMeasurementRepeatability

Open the latest text.log file as shown in the Figure 36 and check the value of the “Mean Z-Height (nm)” and “StdDev Z-Height (nm)” whether those values are within the range listed in Table 6.

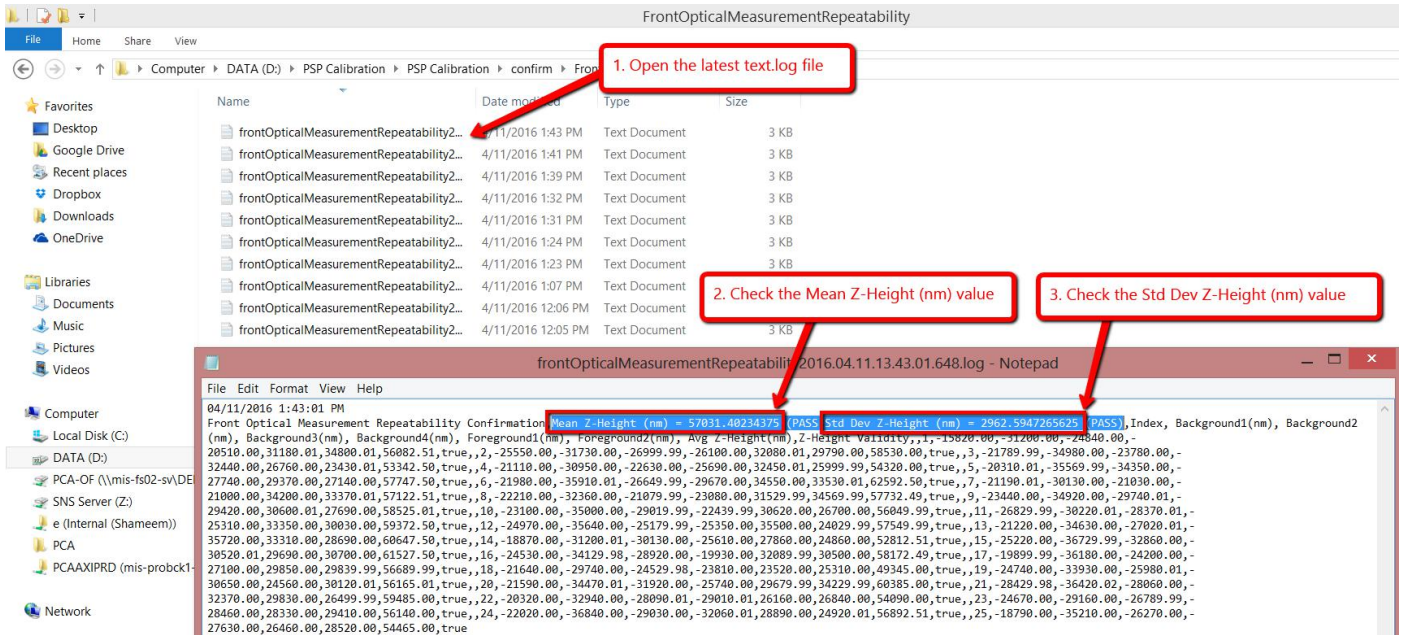


Figure 48 Repeatability test

Log files	Low Limit	High Limit
Mean Z-Height (nm)	20000.0	80000.0
StdDev Z-Height (nm)	0.0	5000.0

Table 6 50 Micron repeatability test

35. At **Confirmation, Diagnostics and Adjustment (CD&A)** , under the **Manual Optical Confirmation**, right click **Rear Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 49 below.

Remarks: The first run will mainly for 500 micron test.

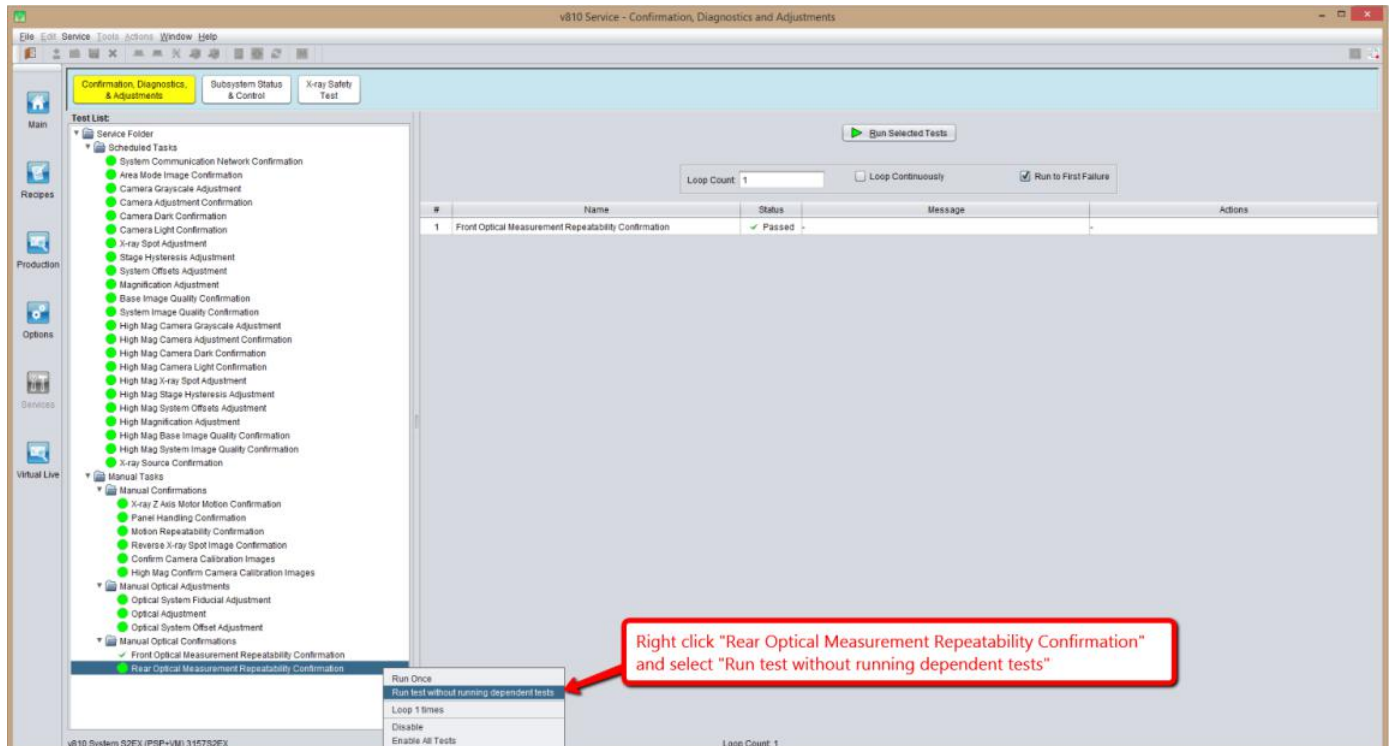


Figure 49 Rear Optical Measurement Repeatability Confirmation

36. Insert the jig once the left outer barrier open as shown in the Figure 50.



Figure 50 Jig loading

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 STD Automated PSP 2 Adjustment Procedure (Manual Load Jig)	Effective Date	2 Oct 2016

37. Select the points of the adjustment dot as shown in the Figure 51.

Remarks: Click to choose the point.

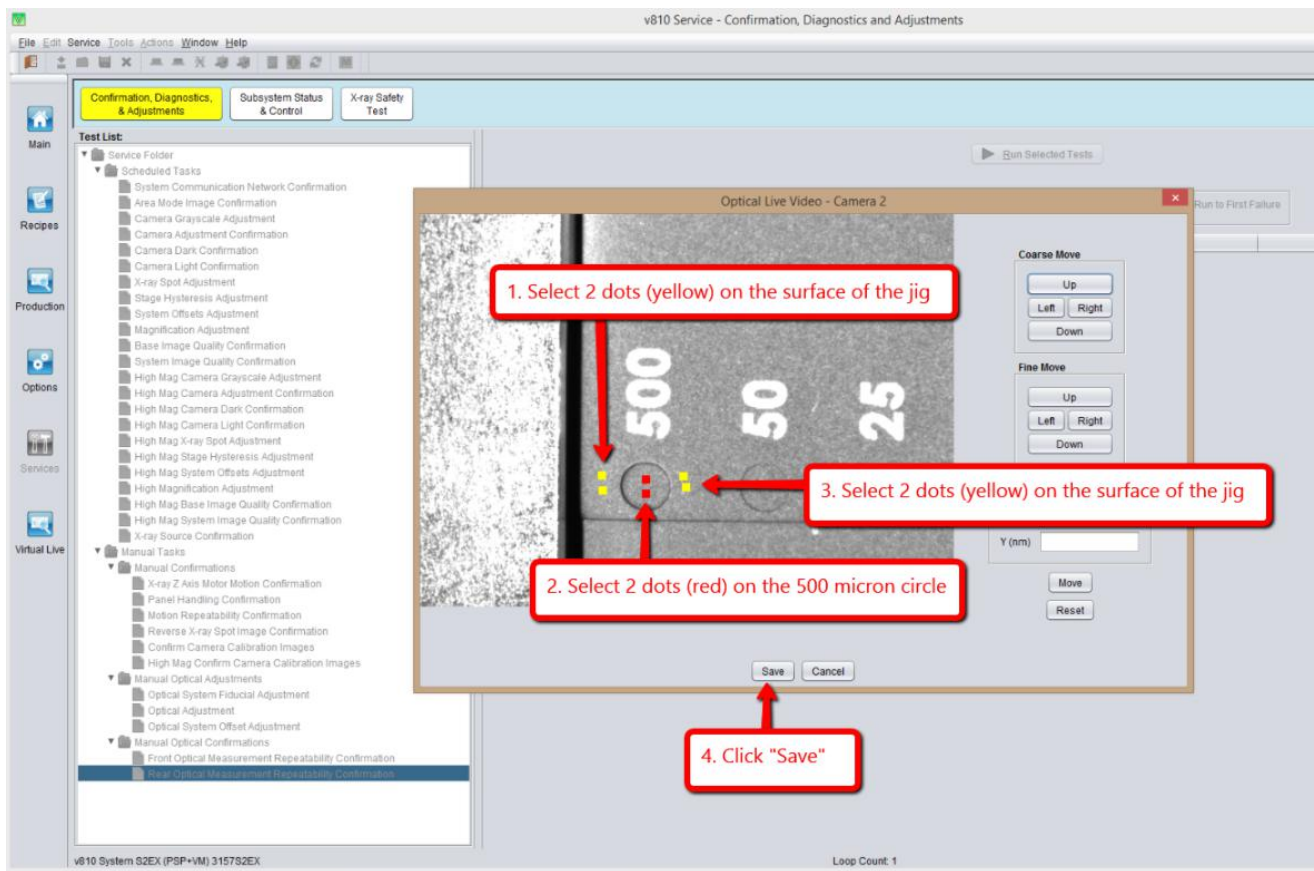


Figure 51 Adjustment Point

38. Once the outer barrier open as shown in the Figure 52, remove the on rail jig from the machine.

After the outer barrier close, the GUI will show the test have passed or failed as shown in the Figure 53.



Figure 52 Jig Removal

	Work Instruction	Doc. Revision	02
	AXI V810 STD Automated PSP 2 Adjustment Procedure (Manual Load Jig)	Effective Date	2 Oct 2016

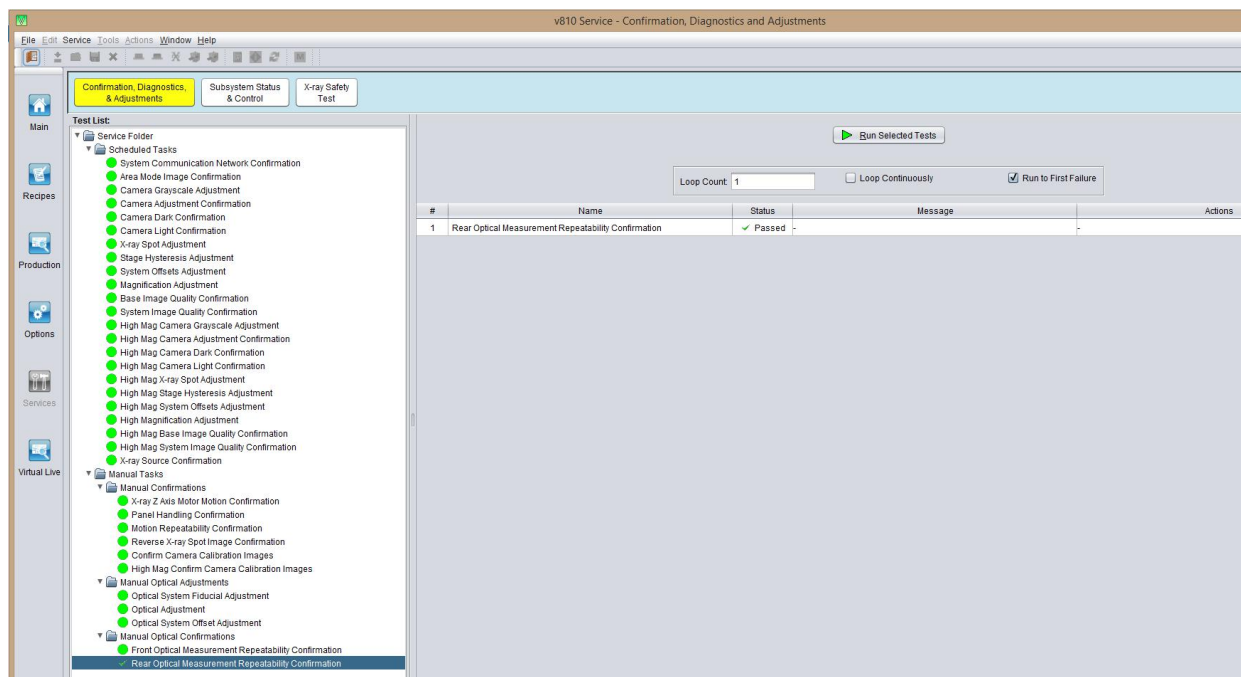


Figure 53 Front Optical Measurement Repeatability Confirmation Passed

39. Go to the following directory

Directory: C:\Program Files\AXI

System\v810\5.0\log\confirm\RearOpticalMeasurementRepeatability

Open the latest text.log file as shown in the Figure 54 and check the value of the “Mean Z-Height (nm)” and “StdDev Z-Height (nm)” whether those values are within the range listed in Table 7.

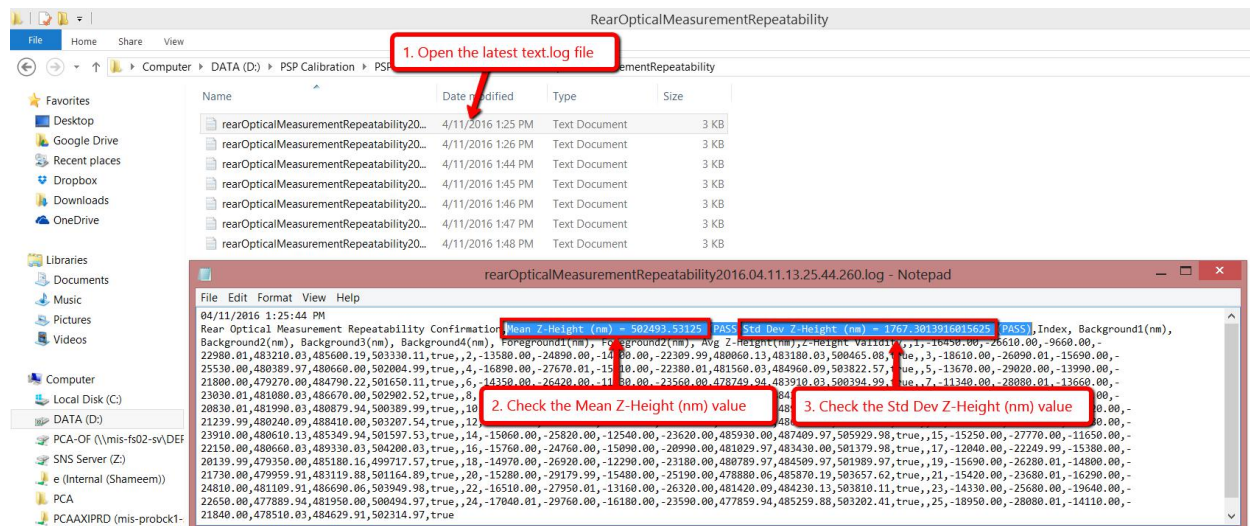


Figure 54 Repeatability test

Log files	Low Limit	High Limit
Mean Z-Height (nm)	400000.0	600000.0
StdDev Z-Height (nm)	0.0	5000.0

Table 7 500 Micron repeatability test

	Work Instruction	Doc. Revision	02
	AXI V810 STD Automated PSP 2 Adjustment Procedure (Manual Load Jig)	Effective Date	2 Oct 2016

40. At **Confirmation, Diagnostics and Adjustment (CD&A)**, under the **Manual Optical Confirmation**, right click **Rear Optical Measurement Repeatability Confirmation** and select **Run test without dependent tests** as shown in the Figure 55 below.

Remarks: The second will run for 50 micron.

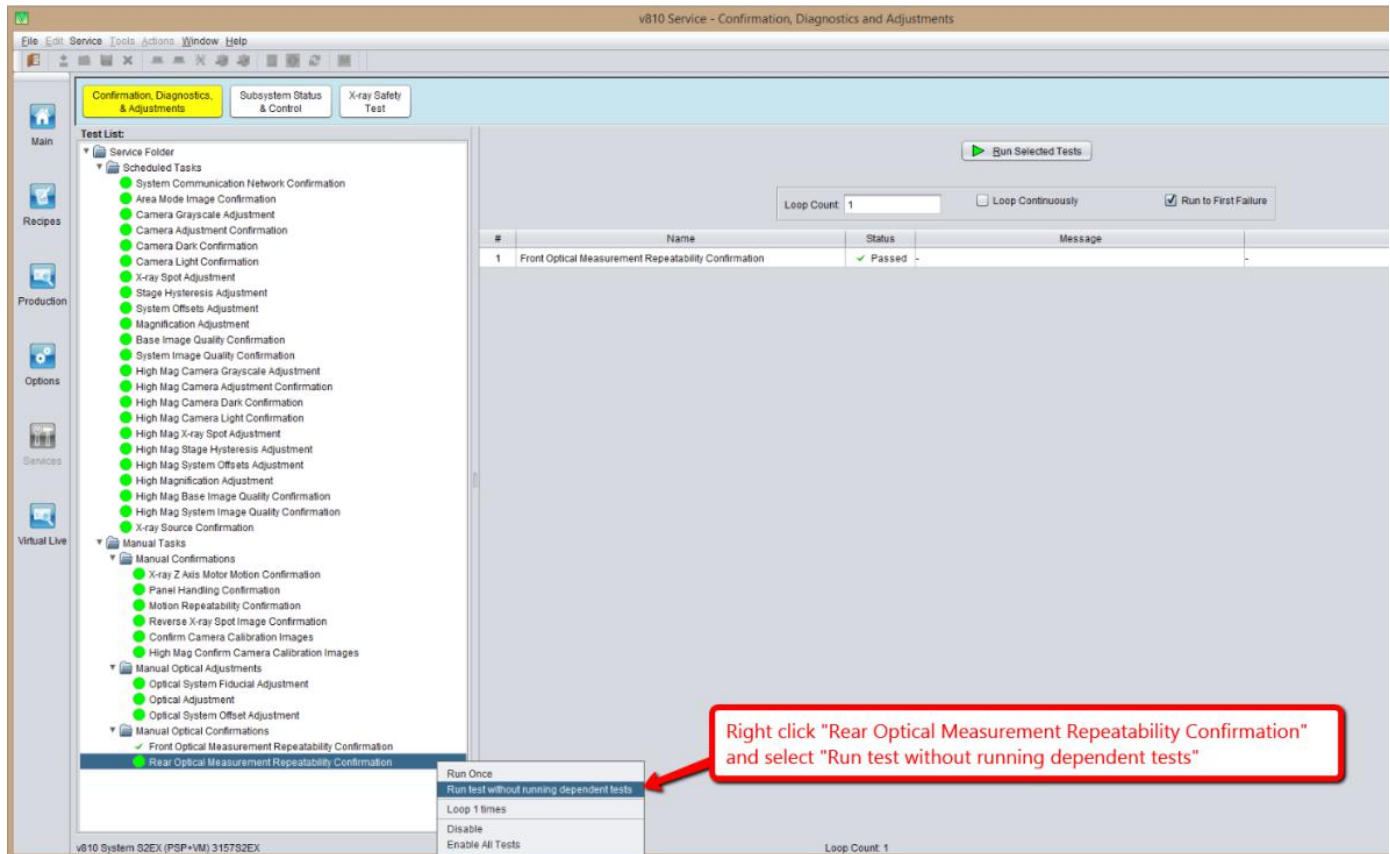


Figure 55 Repeatability test 50 micron

41. Insert the jig once the left outer barrier open as shown in the Figure 56. Once the jig have been loaded, continue with the Step 42.



Figure 56 Jig loading

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 STD Automated PSP 2 Adjustment Procedure (Manual Load Jig)	Effective Date	2 Oct 2016

42. Do select the points of the adjustment dot as shown in the Figure 57.

Remarks: Click to choose the point.

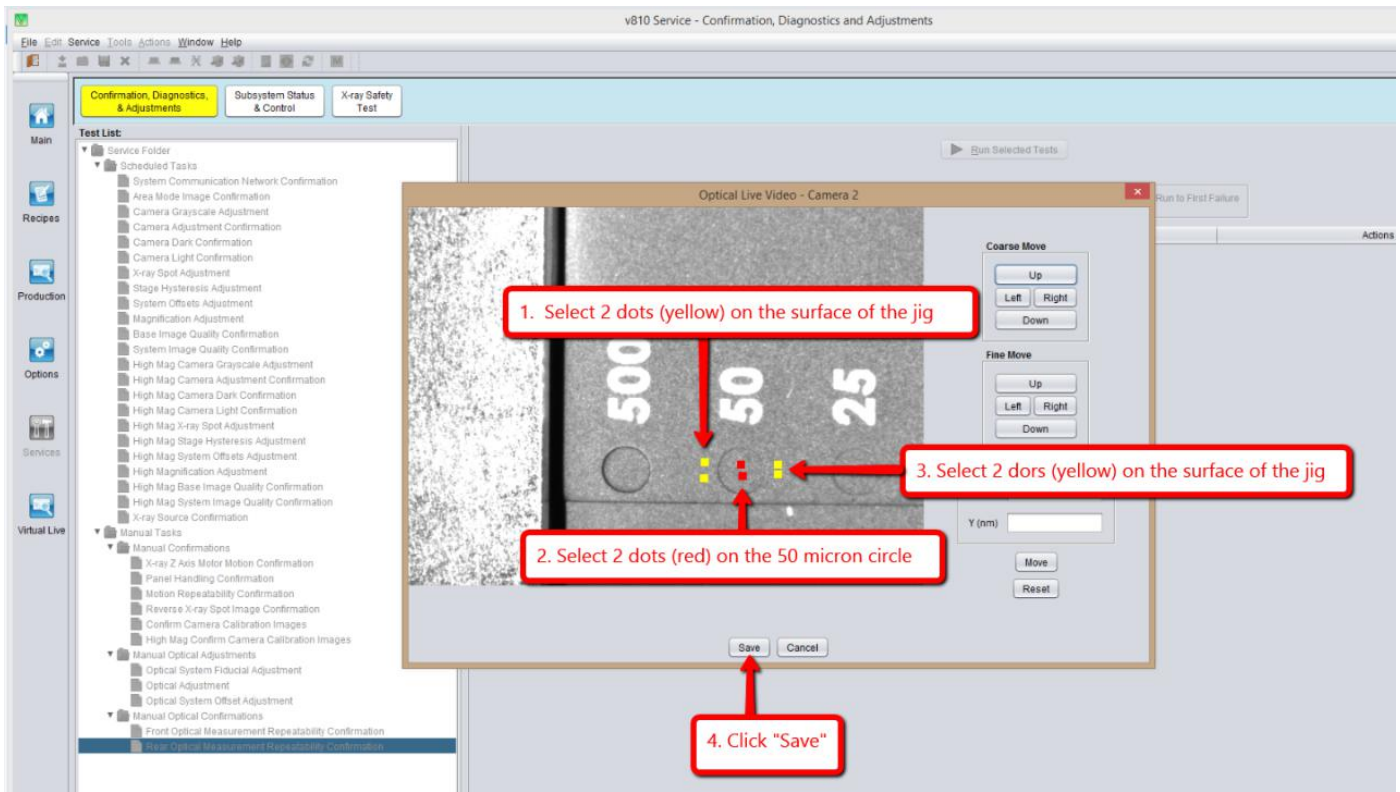


Figure 57 Adjustment Point

43. Once outer barrier open as shown in the Figure 58, remove the on rail jig from the machine. After the outer barrier close, the GUI will show the test have passed or failed as shown in the Figure 59.



Figure 58 Jig Removal

	Work Instruction		Doc. Revision	02
	AXI V810 STD Automated PSP 2 Adjustment Procedure (Manual Load Jig)		Effective Date	2 Oct 2016

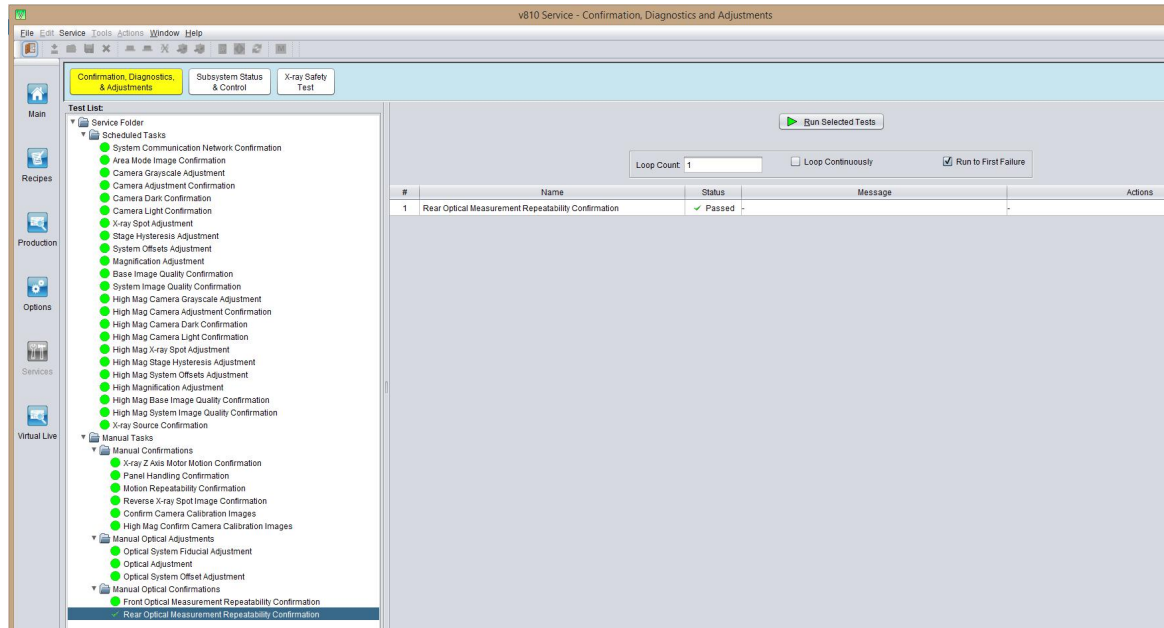


Figure 59 Front Optical Measurement Repeatability Confirmation Passed

44. Go to the following directory

Directory: C:\Program Files\AXI

System\v810\5.0\log\confirm\RearOpticalMeasurementRepeatability

Open the latest text.log file as shown in the Figure 60 and check the value of the “Mean Z-Height (nm)” and “StdDev Z-Height (nm)” whether those values are within the range listed in Table 8.

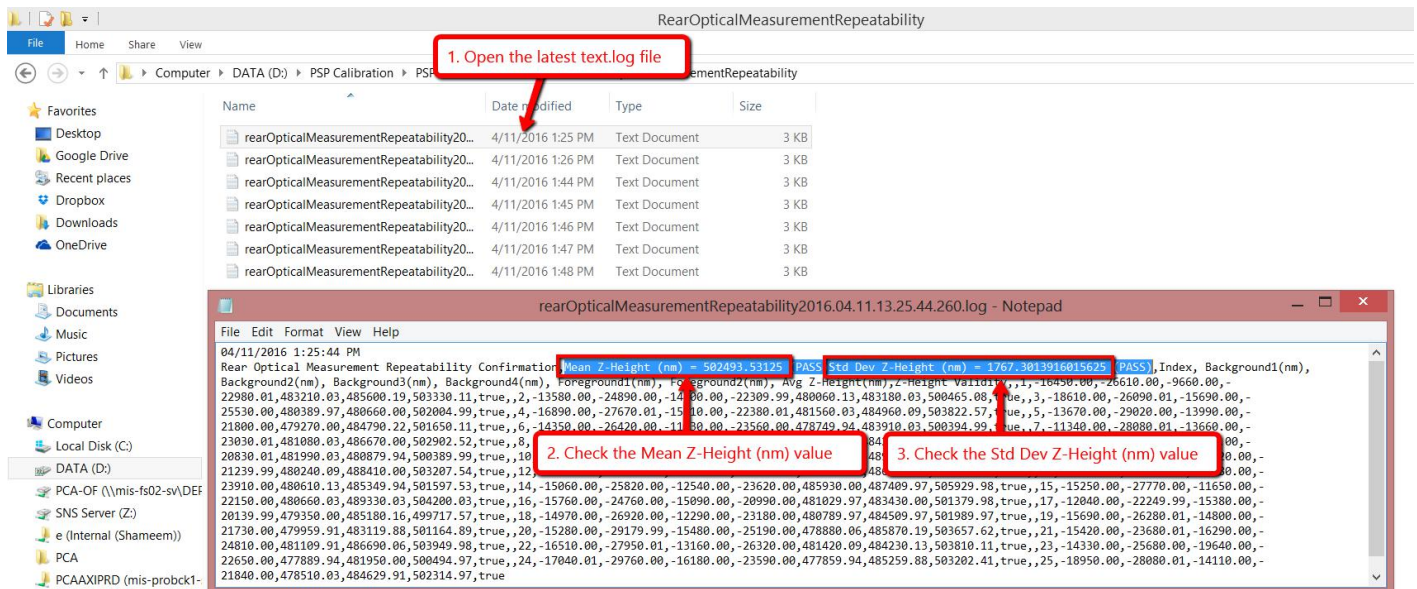


Figure 60 Repeatability test

Log files	Low Limit	High Limit
Mean Z-Height (nm)	400000.0	600000.0
StdDev Z-Height (nm)	0.0	5000.0

Table 8 500 Micron repeatability test

	Work Instruction	Doc. Revision	02
	AXI V810 STD Automated PSP 2 Adjustment Procedure (Manual Load Jig)	Effective Date	2 Oct 2016

45. Send the following folder to Vitrox AXI SNS for further verification,

- i. OpticalCalibration (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\calib

- ii. OpticalSystemOffsetAdjustment (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\calib

- iii. FrontOpticalMeasurementRepeatability (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\confirm

- iv. RearOpticalMeasurementRepeatability (whole folder)

Directory: C:\Program Files\AXI System\v810\5.0\log\confirm